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1 Why we estimate this equation: the HSA Phillips curve

Our estimating equation is the theory-motivated counterpart of the New Keynesian Phillips curve derived by Fujiwara–Matsuyama (2026, *Journal of Monetary Economics*) under the HSA demand system. This section states their result, maps it to what we take to data, and makes explicit how far the empirical specification departs from the theory.

1.1 The HSA Phillips curve

A final-good producer aggregates a continuum of varieties through an HSA demand system with market-share function $s(z)$, where $z = p/A$ is a firm’s price normalized by the single price aggregator A (the reference price faced by firms, an inverse index of competitive pressure). Intermediate firms set prices under a Rotemberg adjustment cost χ and enter endogenously with a one-period time-to-build lag. Solving the firm’s problem, PPI inflation follows (their Eq. 21)

$$\hat{\pi}_t = \beta(1 - \delta) \mathbb{E}_t \hat{\pi}_{t+1} + \underbrace{\frac{\zeta(z) - 1}{\chi}}_{\text{slope on marginal cost}} (\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t) - \underbrace{\frac{1}{\chi} \frac{1 - \rho(z)}{\rho(z)}}_{\text{entry / cost-push}} \hat{N}_t, \quad (1)$$

where $\hat{N}_t$ is the inverse HHI (number of firms), $\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t$ is the inverse markup (the model’s measure of real marginal cost), $\zeta(z) = 1 - s'(z)z/s(z) > 1$ is the price-elasticity function (so the flexible-price markup is $\mu = \zeta/(\zeta - 1)$), and $\rho(z)$ is the pass-through rate.

1.2 Two ways competition enters

(1) The slope is a function of the firm count (structural flattening). The coefficient on marginal cost, $(\zeta(z) - 1)/\chi$, depends on the price elasticity $\zeta(z)$, which through $z = p/A$ and the symmetric-equilibrium condition $s(z) = 1/N$ is a function of the firm count N . Rising concentration—*fewer firms*, higher markups, lower pass-through $\rho(z)$ —lowers this slope, so the Phillips curve flattens *structurally*. Linearizing the slope in the trend level N , $\bar{N}_t$ of the firm count gives

$$\kappa_t = \kappa_0 + \delta \bar{N}_t, \quad (2)$$

so a *positive* δ is the theory’s flattening prediction: the slope moves with the firm count and falls as concentration rises and N declines.

(2) Endogenous entry is a cost-push term (estimation bias). The second term $-\frac{1}{\chi} \frac{1 - \rho(z)}{\rho(z)} \hat{N}_t$ appears only away from CES. It means that the cyclical component of firm entry acts as a cost-push shock through strategic complementarity. Omitting it biases an empirical Phillips-curve estimate (omitted-variable problem). We retain the cyclical term explicitly with its own coefficient $\theta_t = \theta_0 + \gamma \bar{N}_t$.

CES is the irrelevant case. Under CES, $s(z) = \gamma_{\text{CES}} z^{1-\theta}$, so $\zeta(z) = \theta$ is constant and pass-through is $\rho(z) = 1$. The entry term vanishes and the slope is the constant $(\theta - 1)/\chi$: market concentration neither flattens the curve nor moves inflation. This is exactly our CES benchmark, and departing from CES (the HSA models) is what can make δ and the $\hat{N}_t$ term nonzero.

1.3 From theory to the estimating equation

Writing the activity/inverse-markup gap as x_t , adding a backward-looking inertia term (α) and a reduced-form shock e_t , and decomposing the observed firm count into trend and cycle,

$$N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t, \quad (3)$$

where $\bar{N}_t$ is a random-walk-with-drift trend, $\hat{N}_t$ is a stationary AR(2) gap (truncated to the stationary region), and $\nu_t \sim \mathcal{N}(0, \sigma_N^2)$ is measurement error, the HSA Phillips curve becomes the equation we estimate:

$$\boxed{\pi_t = \alpha \pi_{t-1} + (1 - \alpha) \mathbb{E}_t \pi_{t+1} + \kappa_t x_t - \theta_t \hat{N}_t + e_t, \quad \kappa_t = \kappa_0 + \delta \bar{N}_t, \quad \theta_t = \theta_0 + \gamma \bar{N}_t.} \quad (4)$$

The trend $\bar{N}_t$ carries the structural-flattening channel (through δ); the cycle $\hat{N}_t$ carries the entry cost-push channel (through θ_t).

1.4 Distance from the theory

We call this exercise *semi-structural* rather than structural because the specification departs from (1) in several ways. To avoid over-reading the coefficients, we state each:

- the theory is about *PPI* inflation, but there is no PPI expectation series, so we pair observed PPI with a common external expectation series and the expectation term is not a matched structural object;
- the theoretical regressor is real marginal cost / the inverse markup, but the main results use the negative unemployment gap.
- we impose no cross-coefficient restrictions from the theory and approximate κ_t as a *linear* function of $\bar{N}_t$;
- we add backward-looking inertia (α), which is absent from (1);
- expected inflation is an external survey measure (Section 2);

2 Data

Sample. Quarterly, 1982Q1–2012Q4 ($T = 124$) in the US.

Inflation. We compare headline CPI, core CPI, and PPI. Expected inflation $\mathbb{E}\pi$ is the same external survey series in all specifications. Because the Survey of Professional Forecasters does not publish a PPI forecast, the PPI specifications use the same expectation series as CPI. The PPI estimates are therefore not a structural estimate matched through the expectation term, but a semi-structural external-validity check that changes only the observed inflation index. A measurement mismatch with the expectation series also remains for core CPI.

Activity gaps (five). Negative unemployment gap, BN output gap, HP output gap ($100 \times \log$ real output HP-filtered, in the same 100-log-point units as BN), HP labor-share gap, and inverse markup. The unemployment measure enters as the *negative* unemployment gap $x_t = -(u_t - u_t^*)$ —labor-market tightness, positive in expansions and negative in downturns (e.g. $x_t \approx -4$ in 2009), co-moving positively with the output gaps. A positive slope κ on x_t is therefore a conventional downward-sloping Phillips curve.

Competition proxy, and why the main results are mixed-frequency. The competition series is Gustavo’s firm-count measure, i.e. the inverse of the *HHI of U.S. listed firms*. It is **annual**: over 1982–2012 there are 31 observations, not 124.

The main results therefore use a *mixed-frequency* observation scheme. The annual value enters the state-space model in that year’s Q4 and the intervening 93 quarters are treated as *missing*, so the sub-annual firm count is inferred rather than assumed (Section 3.3). Alongside it we report the conventional alternative, in which the annual series is PCHIP-interpolated to a quarterly value and that interpolated value is then used *as data* in all 124 quarters (Section 7).

Earlier revisions of this paper took the interpolated version as the main result. We reversed that after finding that the interpolation is not innocuous. It manufactures within-year variation that does not exist, drives the estimated firm-count measurement error down to $\sigma_N \approx 0.023$ (versus 0.068 under mixed frequency), and through that pins $\bar{N}_t + \hat{N}_t$ so tightly at all 124 quarters that only the *split* between trend and cycle is left free. The consequence is a near-exact posterior ridge between the two states—posterior correlation -0.9996 under interpolation against $+0.13$ under mixed frequency—which is the identification pathology described in Section 8. The interpolated design also forces the AR(2) cycle to a near-unit root and hence to an 82-quarter “cycle” that is a second trend rather than a cycle. None of this is visible if the interpolated series is simply taken as data.

The reassuring part is that δ itself barely moves between the two designs, so the substantive result does not hinge on the choice; what changes is how much of the reported precision is real.

Unit convention. The default transform is $(100 \log N - \overline{100 \log N})/10$, so one unit of $\bar{N}$ or $\hat{N}$ is a ten-log-point change around the sample mean. The reported $\delta, \theta, \theta_0, \gamma$ are all on this ten-log-point scale.

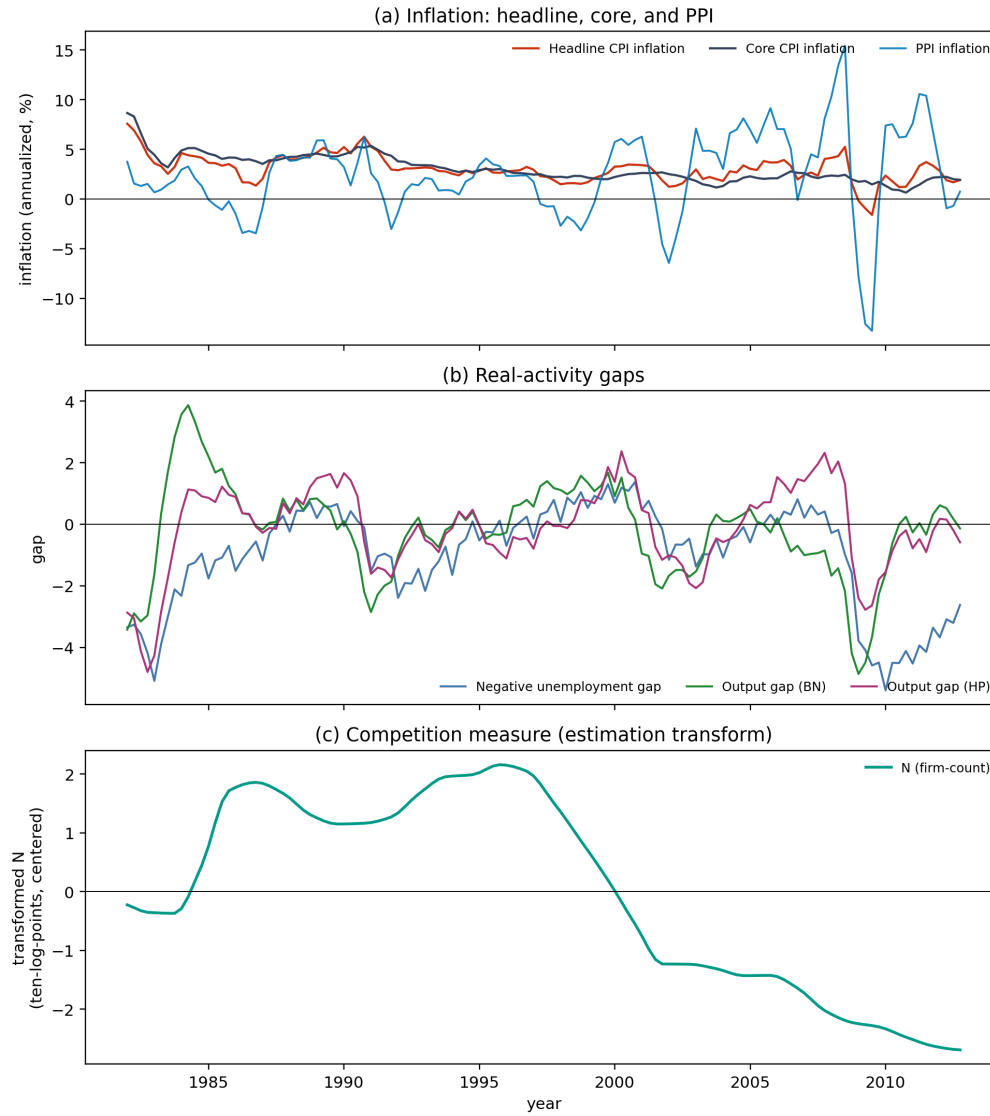


Figure 1: Data. (a) Headline, core, and PPI inflation. (b) Real-activity gaps (negative unemployment gap, BN and HP output gaps). (c) Competition measure (estimation transform of the listed-firm count).

3 Econometric model and estimation

3.1 Model variants

The five models differ only in the restriction on the slope (Table 1).

Table 1: Model variants.

Model	κ_t	θ_t	Role	State block
CES	constant	0	Textbook NKPC benchmark	Gibbs (conjugate)
HSA steady	$\kappa_0 + \delta \bar{N}_t$	0	$\bar{N}$ channel only	Gibbs + exact joint FFBS
HSA dynamic	constant	constant	$\hat{N}$ channel only	Gibbs + exact joint FFBS
HSA const-theta	$\kappa_0 + \delta \bar{N}_t$	constant	HSA full with $\gamma = 0$	Gibbs + exact joint FFBS
HSA full	$\kappa_0 + \delta \bar{N}_t$	$\theta_0 + \gamma \bar{N}_t$	HSA Full	Gibbs + Particle Gibbs

State blocks. The bilinear term $-\gamma \bar{N}_t \hat{N}_t$ makes the joint firm-count state non-linear-Gaussian, so **HSA full is estimated by a Particle Gibbs sampler**: one conditional sequential-Monte-Carlo sweep updates the whole state path (512 particles). This applies to *every* HSA-full cell, under both the PCHIP and the annual-Q4 observation scheme, so the PCHIP-versus-annual-Q4 comparison of Section 7 differs only in the observation scheme and not in the sampler. The table builder refuses to emit a table in which any HSA-full cell was drawn by a different sampler.

Every other model draws the whole state $s_t = (\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)'$ jointly by exact Kalman/FFBS. That includes **HSA const-theta**: with $\gamma = 0$ the inflation observation is linear in the joint state, so the model is linear-Gaussian and admits an exact joint draw. Earlier revisions sampled it with the same two alternating blocks as HSA full; because $\bar{N}_t$ and $\hat{N}_t$ are almost perfectly negatively correlated in the posterior (about -0.999 : the firm-count equation pins their sum but barely their split), that blocking moved their common level with an integrated autocorrelation time of order 10^3 sweeps and did not mix. This is a purely computational change—it removes a sampler pathology and creates no identification that the data do not contain. Coefficients, variances, priors, chains, iterations, burn-in and thinning are unchanged throughout.

3.2 The inflation equation actually estimated

To put the theoretical equation (4) directly in regression form, we move expected inflation to the left-hand side and define

$$y_t \equiv \pi_t - E_t \pi_{t+1}, \quad a_t \equiv \pi_{t-1} - E_t \pi_{t+1} \quad (5)$$

Making the unforecastable component of the activity variable explicit, the estimating system is

$$y_t = \alpha a_t + (\kappa_0 + \delta \bar{N}_t) x_t - (\theta_0 + \gamma \bar{N}_t) \hat{N}_t + \lambda_{e\zeta} \zeta_t + \eta_t, \quad (6)$$

$$x_t = \phi_1 x_{t-1} + \zeta_t, \quad \eta_t \perp \zeta_t. \quad (7)$$

That is, the original NKPC shock decomposes as $e_t = \lambda_{e\zeta}\zeta_t + \eta_t$. Estimating $\lambda_{e\zeta}$ allows a correlation between the contemporaneous activity shock and the inflation shock, rather than forcing a simple regression that treats x_t as exogenous. The saved correlation coefficient is

$$\rho_{e\zeta} = \frac{\lambda_{e\zeta}\sigma_\zeta}{\sqrt{\lambda_{e\zeta}^2\sigma_\zeta^2 + \sigma_\eta^2}} \quad (8)$$

The main HSA steady sets $\theta_0 = \gamma = 0$; HSA dynamic sets $\delta = \gamma = 0$.

For numerical stability the implementation updates internal coefficients on the rescaled regressors $x_t/100$ and $x_t\bar{N}_t/100$ for κ_0 and δ ; the coefficients are rescaled back by 100 on saving, so the $\kappa_0, \delta, \kappa_t$ in the text, tables, and figures are all in the physical units of (6). They must not be multiplied by 100 or divided by 100 again.

3.3 State-space representation splitting the firm count into trend and cycle

The observation and transition equations for the firm count are

$$N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t, \quad \nu_t \sim N(0, \sigma_N^2), \quad (9)$$

$$\hat{N}_t = \rho_1 \hat{N}_{t-1} + \rho_2 \hat{N}_{t-2} + u_t, \quad u_t \sim N(0, \sigma_u^2), \quad (10)$$

$$\bar{N}_t = n + \bar{N}_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2) \quad (11)$$

where $\bar{N}_t$ is the low-frequency trend with drift n and $\hat{N}_t$ is a stationary AR(2) cycle. Rather than a two-step procedure that first filters the observed firm count into components and then substitutes them into the NKPC, we extract both states using the firm-count equation and the inflation equation within the same Gibbs iteration.

Two observation frequencies; the main results use the first. The firm count is annual, so the *mixed-frequency* design places the annual value in each year's Q4 and treats Q1–Q3 as missing,

$$N_t^{\text{obs}} = \begin{cases} (100 \log N_{y(t)}^{\text{annual}} - \bar{c})/10, & t \in \mathcal{T}_N \text{ (Q4)}, \\ \text{missing}, & t \notin \mathcal{T}_N \text{ (Q1–Q3)}, \end{cases} \quad |\mathcal{T}_N| = 31, \quad (12)$$

where $\bar{c}$ is a fixed centering constant, shared with the interpolated series so the coefficient units are comparable across designs. The alternative *interpolated* design instead PCHIP-interpolates the annual series to quarterly and applies (9) at all 124 quarters; Section 7 reports it and documents what the interpolation costs. In Q4 the Kalman update uses two observation rows—the firm-count equation and the inflation equation—and in Q1–Q3 only the one inflation row. Because the state transition and prediction continue at every quarter, the Q1–Q3 $\bar{N}_t, \hat{N}_t$ are not simple interpolations but FFBS draws conditioned on the annual anchors, the state transition, and the inflation equation, updated every iteration. The inverse-gamma update of σ_N^2 likewise uses only the 31 finite Q4 residuals.

For the main HSA steady the state is

$$s_t = (\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)', \quad s_t = c + F s_{t-1} + w_t \quad (13)$$

with

$$c = \begin{bmatrix} 0 \\ 0 \\ n \end{bmatrix}, \quad F = \begin{bmatrix} \rho_1 & \rho_2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad Q = \text{Var}(w_t) = \begin{bmatrix} \sigma_u^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma_\varepsilon^2 \end{bmatrix}. \quad (14)$$

Stacking the firm-count observation with the inflation equation net of its known part,

$$y_t^\pi = y_t - \alpha a_t - \kappa_0 x_t - \lambda_{e\zeta} \zeta_t \quad (15)$$

gives

$$\underbrace{\begin{bmatrix} N_t^{\text{obs}} \\ y_t^\pi \end{bmatrix}}_{z_t} = \underbrace{\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & \delta x_t \end{bmatrix}}_{H_t} s_t + v_t, \quad \text{Var}(v_t) = \begin{bmatrix} \sigma_N^2 & 0 \\ 0 & \sigma_\eta^2 \end{bmatrix}. \quad (16)$$

Conditional on the coefficients and variances this is a linear-Gaussian state-space model, so $s_{0:T}$ can be drawn jointly by an exact Kalman filter / forward-filtering backward-sampling (FFBS).

The initial state starts from the fixed prior

$$s_0 = (\hat{N}_0, \hat{N}_{-1}, \bar{N}_0)' \sim N(m_0, P_0), \quad m_0 = (0, 0, 0)', \quad P_0 = 10I_3 \quad (17)$$

We do not substitute N_0^{obs} into m_0 ; the period-0 firm count and inflation enter the ordinary Kalman observation update. The first AR(2) transition also actually uses the prior-updated $\hat{N}_{-1}$. This avoids double-using the initial observation or dropping the first AR(2) likelihood.

3.4 Conditional posteriors of the coefficients and variances

Take the main model as an example, with $\beta = (\alpha, \kappa_0, \delta)'$, $X_t = (a_t, x_t, x_t \bar{N}_t)$, and $y_t^* = y_t - \lambda_{e\zeta} \zeta_t$. Under the independent normal prior $\beta \sim N(b_0, V_0)$, we update jointly from

$$V_1 = \left(V_0^{-1} + \frac{X'X}{\sigma_\eta^2} \right)^{-1}, \quad (18)$$

$$b_1 = V_1 \left(V_0^{-1} b_0 + \frac{X' y^*}{\sigma_\eta^2} \right), \quad (19)$$

$$\beta \mid \text{rest} \sim N(b_1, V_1) \quad (20)$$

The implementation uses the 100-rescaled X and coefficients noted above, but the formula is identical. The baseline estimation imposes no sign restriction, so both δ and the end-of-sample κ_t can be negative.

We update $\lambda_{e\zeta}$ from the normal conditional posterior of the regression of e_t on ζ_t . We update ϕ_1 using not only (7) but also the information in $e_t = \lambda_{e\zeta}(x_t - \phi_1 x_{t-1}) + \eta_t$. We draw ρ_1, ρ_2 from a normal conditional posterior truncated to the AR(2) stationary region,

$$|\rho_2| < 1, \quad \rho_1 + \rho_2 < 1, \quad \rho_2 - \rho_1 < 1 \quad (21)$$

The fallback count and rejection rate when no proposal is accepted within 2,000 draws are saved to each run’s metadata.

The variance priors are $\sigma^2 \sim \text{IG}(a, b)$ (density kernel $(\sigma^2)^{-(a+1)} \exp(-b/\sigma^2)$), updated from the inverse-gamma conditional posterior that adds the residual sum of squares. The concrete baseline priors are in Table 2.

Table 2: Baseline priors for the main HSA steady model. Normal distributions show mean and standard deviation; inverse-gamma distributions show (a, b) . The coefficient priors are common to the other models.

Block	Parameter	Prior	Interpretation
Inflation eq.	α	$N(0.5, 0.2^2)$	Backward inertia
	κ, κ_0	$N(0.1, 0.2^2)$	Slope level
	δ	$N(0, 0.02^2)$	Slope dependence on firm count
	θ, θ_0	$N(0.1, 0.2^2)$	Cyclical entry effect
	γ	$N(0, 0.02^2)$	Time-varying coefficient of entry
Activity eq.	ϕ_1	$N(0.7, 0.2^2)$	AR(1) of the activity variable
	$\lambda_{e\zeta}$	$N(0, 0.5^2)$	Simultaneous-shock correction
Firm-count state	(ρ_1, ρ_2)	$N(0.5, 0.2^2) \times N(-0.5, 0.2^2)$	Truncated to stationary region
	n	$N(0, 0.1^2)$	Trend drift
Variances	$\sigma_\eta^2, \sigma_\zeta^2$	$\text{IG}(2, 2), \text{IG}(.001, .001)$	Inflation / activity shock
	$\sigma_u^2, \sigma_\varepsilon^2, \sigma_N^2$	$\text{IG}(2, .02), \text{IG}(2, .01), \text{IG}(2, .01)$	State / measurement error

For HSA dynamic the shock covariance takes $\Sigma \sim \text{IW}(8, S_0)$ with $S_0 = \text{diag}(3, 3, 0.06, 0.03)$. Under the default (e, ζ) -only, the corresponding marginals of this prior apply to the (e, ζ) block and the two state variances.

3.5 What the Kalman filter and FFBS do

Write the predicted mean and variance at time t as $a_t^s, P_{t|t-1}$ and the filtered mean and variance as m_t, P_t . The forward update for (14)–(16) is

$$a_t^s = c + Fm_{t-1}, \quad P_{t|t-1} = FP_{t-1}F' + Q, \quad (22)$$

$$q_t = z_t - H_t a_t^s, \quad S_t = H_t P_{t|t-1} H_t' + R, \quad (23)$$

$$K_t = P_{t|t-1} H_t' S_t^{-1}, \quad m_t = a_t^s + K_t q_t \quad (24)$$

The covariance update uses the numerically stable Joseph form. After drawing the endpoint $s_T \sim N(m_T, P_T)$, we draw the smoothed state path jointly with

$$J_t = P_t F' P_{t+1|t}^{-1}, \quad (25)$$

$$s_t \mid s_{t+1}, z_{1:T} \sim N(m_t + J_t(s_{t+1} - c - Fm_t), P_t - J_t P_{t+1|t} J_t') \quad (26)$$

for $t = T - 1, \dots, 0$. Thus the plotted $\bar{N}_t, \hat{N}_t, \kappa_t$ carry the posterior uncertainty not only of the coefficients but also of the state decomposition.

3.6 One Gibbs iteration and the differences across models

One iteration of HSA steady is, in order:

1. update $(\alpha, \kappa_0, \delta)$ from the current $\bar{N}_t$ via (20);
2. update $\lambda_{e\zeta}$ and ϕ_1 , and recompute ζ_t, η_t ;
3. update $\sigma_\zeta^2, \sigma_\eta^2$;
4. update (ρ_1, ρ_2) and σ_u^2 ;
5. update n and σ_ε^2 , then σ_N^2 ;
6. update the entire $s_{0:T}$ by FFBS using the updated coefficients and variances.

CES is a conjugate regression benchmark with no latent firm-count state.

HSA dynamic handles the covariance of $r_t = (e_t, \zeta_t, u_t, \varepsilon_t)'$; under the default (e, ζ) -only it updates the 2×2 block of (e_t, ζ_t) and $\sigma_u^2, \sigma_\varepsilon^2$ directly on the constrained parameter space, rather than zeroing the off-diagonal of an unconstrained inverse-Wishart draw after the fact.

HSA full contains the state-by-state product $-\gamma \bar{N}_t \hat{N}_t$, so the joint firm-count state $(\bar{N}_t, \hat{N}_t)$ is not linear-Gaussian at once. HSA full is therefore updated by the joint *Particle Gibbs* step detailed in Section 3.7—a conditional sequential-Monte-Carlo update of the whole state path (512 particles)—which replaces the two alternating exact-FFBS blocks used previously. HSA const-theta ($\gamma = 0$) is a different case: without the bilinear term the inflation observation loads linearly on the joint state, $[-\theta_0, 0, \delta x_t] s_t$, so the model is linear-Gaussian and the whole path is drawn in a single exact FFBS sweep—the same routine HSA steady uses, of which HSA steady is the $\theta_0 = 0$ case. In both models the weak signal of the cyclical state $\hat{N}_t$, which must be identified as a separate coefficient, keeps identification weaker than for the other models; neither the joint Particle Gibbs update nor the exact joint FFBS removes that weak identification, and neither is claimed to.

3.7 The Particle Gibbs state update (HSA full)

For HSA full the bilinear term $-\gamma \bar{N}_t \hat{N}_t$ in (6) makes the joint state $s_t = (\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)'$ non-linear-Gaussian, so the exact FFBS blocks used above do not apply jointly. We instead draw the whole path $s_{0:T}$ with one *Particle Gibbs* sweep—a conditional sequential-Monte-Carlo step that leaves the exact conditional $p(s_{0:T} \mid \theta, y, N^{\text{obs}})$ invariant for any number of particles P (Andrieu, Doucet and Holenstein, 2010).

Proposal and weights. Particles are propagated from the state transition (10)–(11) (a bootstrap proposal): given the ancestor $s_{t-1}^{(i)}$,

$$\hat{N}_t^{(i)} = \rho_1 \hat{N}_{t-1}^{(i)} + \rho_2 \hat{N}_{t-2}^{(i)} + u_t^{(i)}, \quad \bar{N}_t^{(i)} = n + \bar{N}_{t-1}^{(i)} + \varepsilon_t^{(i)}, \quad u_t^{(i)} \sim N(0, \sigma_u^2), \quad \varepsilon_t^{(i)} \sim N(0, \sigma_\varepsilon^2). \quad (27)$$

Each particle is weighted by *both* observation equations—the inflation equation (6) and the firm-count measurement (9):

$$\log w_t^{(i)} = -\frac{(y_t - \mu_t^{(i)})^2}{2\sigma_\eta^2} - \frac{(N_t^{\text{obs}} - \bar{N}_t^{(i)} - \hat{N}_t^{(i)})^2}{2\sigma_N^2}, \quad (28)$$

$$\mu_t^{(i)} = \alpha a_t + (\kappa_0 + \delta \bar{N}_t^{(i)})x_t - (\theta_0 + \gamma \bar{N}_t^{(i)})\hat{N}_t^{(i)} + \lambda_{e\zeta}\zeta_t, \quad (29)$$

with $y_t = \pi_t - \mathbb{E}_t \pi_{t+1}$, $a_t = \pi_{t-1} - \mathbb{E}_t \pi_{t+1}$ and $\zeta_t = x_t - \phi_1 x_{t-1}$ (constants common to all particles at t cancel under normalisation). Normalised weights $W_t^{(i)} = w_t^{(i)} / \sum_j w_t^{(j)}$ are formed with a log-sum-exp for stability, and $\text{ESS}_t = 1 / \sum_i (W_t^{(i)})^2$.

Conditional SMC. Let $s_{0:T}^{\text{ref}}$ be the state path from the previous MCMC iteration, pinned to particle slot 1.

1. $t = 0$: set $s_0^{(1)} = s_0^{\text{ref}}$ and draw $s_0^{(i)} \sim N(m_0, P_0)$ for $i \geq 2$ from the initial prior (17); weight by (28).
2. $t = 1, \dots, T$: resample ancestors $a_t^{(i)} \sim \text{Categorical}(W_{t-1})$ for $i \geq 2$ and set $a_t^{(1)} = 1$, so the reference keeps its lineage; propagate $s_t^{(i)}$ from the transition above given $s_{t-1}^{(a_t^{(i)})}$, with $s_t^{(1)} = s_t^{\text{ref}}$; weight by (28). The pair $(\hat{N}_t, \hat{N}_{t-1})$ in s_t preserves the AR(2) genealogy.
3. Draw a terminal index $J \sim \text{Categorical}(W_T)$ and trace its ancestors backward to return one joint path $s_{0:T}$, which is the new state draw and the reference for the next iteration.

3.8 Iterations, saved draws, and convergence criteria

Each cell uses two independent chains, with a *total* of 12,000 iterations per chain. We discard the first 4,000 as burn-in and save every 5th of the remaining 8,000, giving 1,600 draws per chain and 3,200 draws per cell for diagnostics and summaries. The main analysis is 75 cells: the baseline 45 (3 price indices $\times$ 3 activity measures $\times$ 5 models) plus the 30 weak/tight-prior cells. Separately, we estimate 2 inverse-markup CES/HSA-dynamic cells with the same settings.

What “converged” means here. The threshold is maximum $\hat{R} \leq 1.01$ and minimum bulk ESS ≥ 400 , but a single run has several distinct groups of quantities and they do not pass or fail together. We therefore compute and report the rule on three groups separately:

1. **scalar parameters** — every stored scalar: α , the slope terms, the entry terms, ρ_1, ρ_2 , the trend drift n , ϕ_1 , $\lambda_{e\zeta}$ and *every* estimated variance;
2. **latent-state paths** — $\bar{N}_t$ and $\hat{N}_t$, maximum $\hat{R}$ and minimum ESS along the path;
3. **derived paths** — κ_t and, where estimated, θ_t .

The † mark in the coefficient tables is the *scalar* rule. A cell marked “OK (coef)” passes on the coefficients but not on the latent-state paths; “OK” passes on all three. We state this explicitly because the two are routinely conflated: marginal coefficient chains can mix well while the joint posterior including the states has not, and reporting only the former would overstate what has been established. Table 21 gives the group-by-group numbers for every model.

Under the main mixed-frequency design, 60 of the 77 runs fail the scalar rule; under the interpolated design the figures are 38 of 77. The two designs fail for different reasons, and the difference is informative rather than incidental. Under interpolation the worst-mixing scalar is the trend drift n in 35 of the 38 failing cells—the level ridge of Section 8 surfacing in the parameter block. Under mixed frequency it is the AR(2) block (ρ_1, ρ_2) in 53 of 61 cells, which for HSA steady is a nuisance object because $\theta = 0$ keeps $\hat{N}_t$ out of the inflation equation. The parameters the conclusions rest on fail at the *same* rate in the two designs: δ in 8 of 61 cells and κ_0 in 6 of 61, either way. The 95% credible intervals are the 2.5% and 97.5% quantiles of the saved draws; the plotted points are posterior means.

4 Main results

All results in this section use the **mixed-frequency** firm-count design: the 31 annual observations enter in their own Q4 and the 93 intervening quarters are inferred inside the state-space model (Section 3.3). Table 3 gives the main-model (HSA steady) estimates for every price index and activity measure, so each result can be read against the specification it comes from. The main specification—**core CPI, negative unemployment gap**—is in bold; a positive δ means the slope flattens as the firm count falls.

Table 3: Headline results (main model: HSA steady, baseline priors, mixed-frequency annual-Q4, $T = 124$), by price index and activity measure. δ is the competition dependence of the slope (a positive δ means the slope flattens as the firm count falls); BF_{10} is the Savage–Dickey Bayes factor against $\delta = 0$; κ_0 is the slope at average competition; the last column is the implied κ_t from the start to the end of the sample. The **main specification (core CPI, negative unemployment gap)** is in bold. †: fails the *coefficient* convergence rule ($\hat{R} \leq 1.01$ and bulk ESS ≥ 400 over every scalar parameter, including the trend drift n and all variances). “OK (coef)” marks a cell that passes on the coefficients but not on the latent-state paths; the group-by-group diagnostics are in Table 21. The $\delta \hat{R}/\text{ESS}$ column is that coefficient’s own mixing, which the blanket flag does not show.

Activity measure	Price	δ [95% CI]	BF_{10}	κ_0 [95% CI]	κ_t start→end	$\delta \hat{R}/\text{ESS}$	Conv.
Negative unemployment gap	Core CPI	+0.022 [+0.008, +0.038]†	36.2	+0.056 [+0.021, +0.092]	+0.110 → -0.002	1.000 / 3087	watch
	Headline CPI	+0.029 [+0.005, +0.053]†	11.2	+0.075 [+0.009, +0.145]	+0.143 → -0.001	1.000 / 3378	watch
	PPI	+0.009 [-0.028, +0.047]†	1.1	+0.080 [-0.134, +0.296]	+0.093 → +0.055	1.000 / 3277	watch
HP output gap	Core CPI	+0.028 [+0.009, +0.047]†	19.6	+0.073 [+0.022, +0.124]	+0.152 → +0.001	1.001 / 2677	watch
	Headline CPI	-0.006 [-0.039, +0.027]†	0.9	+0.140 [+0.048, +0.231]	+0.135 → +0.155	1.000 / 3130	watch
	PPI	-0.004 [-0.043, +0.035]†	1.0	+0.212 [-0.040, +0.469]	+0.207 → +0.221	1.000 / 3322	watch
BN output gap	Core CPI	+0.000 [-0.031, +0.025]†	0.7	+0.030 [-0.021, +0.081]	+0.041 → +0.030	1.018 / 145	watch
	Headline CPI	-0.019 [-0.052, +0.013]†	1.6	+0.053 [-0.033, +0.139]	+0.037 → +0.103	1.002 / 2618	watch
	PPI	-0.008 [-0.046, +0.031]†	1.0	+0.187 [-0.051, +0.437]	+0.177 → +0.207	1.001 / 2904	watch

Reading the convergence column. Every cell in Table 3 carries a †, and it would be a mistake to read that as “ δ is not estimated reliably here.” The blanket flag is set by whichever scalar mixes worst, and under mixed frequency that is the AR(2) block (ρ_1, ρ_2) in 53 of the 61 HSA cells. For **HSA steady** the AR(2) block governs $\hat{N}_t$, which—because $\theta = 0$ in this model—does not enter the inflation equation at all. The coefficient the conclusions rest on is δ , and its own diagnostics are in the penultimate column: $\hat{R} = 1.000$ with bulk ESS between 2,600 and 3,400 in eight of the nine cells. Section 8 states which failures do and do not bear on the conclusions.

Table 4 then places the main model against the other model variants (CES, HSA dynamic, HSA const-theta, HSA full) for the negative unemployment gap. CES and HSA dynamic hold the slope constant, so they have no δ ; the HSA models with a firm-count-dependent slope estimate δ , and HSA steady is the main model. Full coefficients (θ , γ , and the κ_t path) for every activity measure are in Appendix C.

Table 4: Model comparison for the negative unemployment gap (baseline priors, mixed-frequency annual-Q4), by price index. Slope is κ for CES / HSA dynamic and κ_0 otherwise; the competition dependence δ (and its BF_{10}) is defined only for the HSA models with a firm-count-dependent slope. The main model, **HSA steady**, is in bold; full coefficients (θ , γ , κ_t path) for every activity measure are in Appendix C. State blocks: CES conjugate; HSA steady / HSA dynamic exact joint FFBS; HSA const-theta joint FFBS; HSA full Particle Gibbs. †: fails the coefficient convergence rule; “OK (coef)” passes on coefficients but not on the latent-state paths.

Model	Price	slope [95% CI]	δ [95% CI]	$\text{BF}_{10}(\delta)$	Conv.
HSA steady	Core CPI	+0.056 [+0.021, +0.092]	+0.022 [+0.008, +0.038] [†]	36.2	watch
	Headline CPI	+0.075 [+0.009, +0.145]	+0.029 [+0.005, +0.053] [†]	11.2	watch
	PPI	+0.080 [-0.134, +0.296]	+0.009 [-0.028, +0.047] [†]	1.1	watch
HSA dynamic	Core CPI	+0.040 [+0.009, +0.071]	–	–	watch
	Headline CPI	+0.055 [-0.011, +0.117]	–	–	watch
	PPI	+0.051 [-0.155, +0.263]	–	–	watch
HSA const-theta	Core CPI	+0.057 [+0.020, +0.095]	+0.022 [+0.004, +0.038] [†]	21.9	watch
	Headline CPI	+0.079 [+0.011, +0.152]	+0.029 [+0.003, +0.055] [†]	7.6	watch
	PPI	+0.079 [-0.129, +0.289]	+0.009 [-0.030, +0.047] [†]	1.1	watch
HSA full	Core CPI	+0.064 [+0.022, +0.109]	+0.025 [+0.008, +0.043] [†]	19.6	watch
	Headline CPI	+0.080 [+0.011, +0.152]	+0.029 [+0.002, +0.054] [†]	6.8	watch
	PPI	+0.075 [-0.140, +0.284]	+0.010 [-0.029, +0.047] [†]	1.1	watch

4.1 The negative-unemployment-gap slope flattens as the firm count falls

For the baseline HSA steady on the main sample, the core-CPI negative-unemployment-gap specification gives

$$\delta = +0.022[+0.008, +0.038],$$

and the Savage–Dickey Bayes factor against $\delta = 0$ is $\text{BF}_{10} = 36.2$. Because $\bar{N}_t$ falls over the sample, the implied slope κ_t declines monotonically from +0.110 to -0.002 : as the firm count falls (concentration rises), the Phillips curve flattens, the sign the HSA mechanism predicts. Headline CPI also gives $\delta = +0.029[+0.005, +0.053]$ ($\text{BF}_{10} = 11.2$), so the CPI result for the negative unemployment gap is robust to the inflation index. Figure 2 plots the implied κ_t path; its end-of-sample band includes zero, so the accurate statement is a flattening to near zero rather than a strictly positive residual slope.

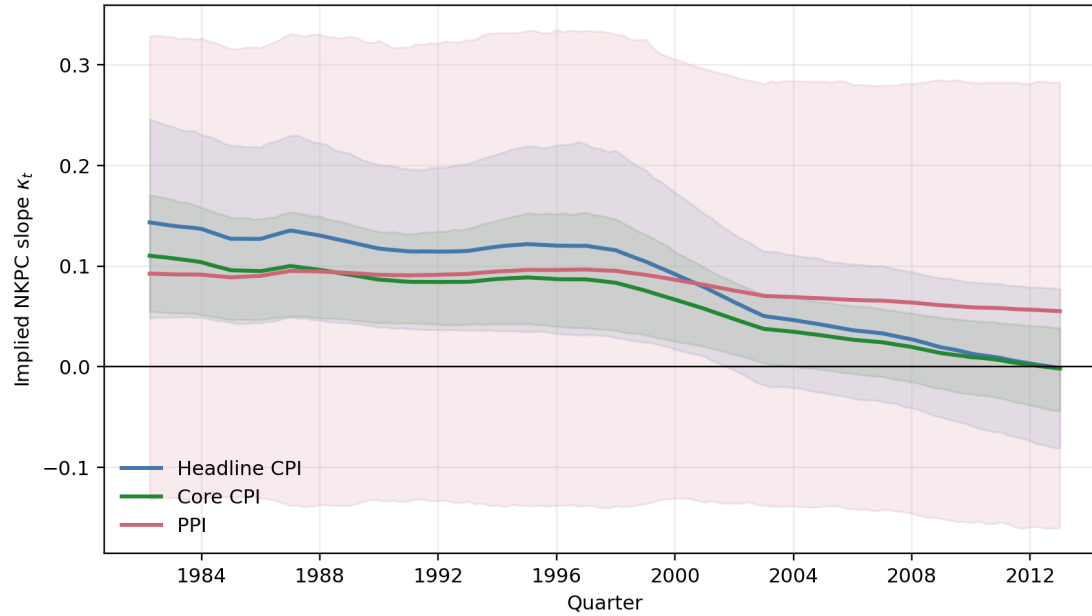


Figure 2: Time-varying slope κ_t for the negative unemployment gap under the mixed-frequency design (posterior mean and 95% band). The band includes the uncertainty from inferring the 93 unobserved quarterly firm counts. Under CPI it declines to near zero by the end of the sample, whereas the PPI band is wide.

The estimated Phillips-curve slope in the core-CPI specification declines from about 0.11 at the beginning of the sample to approximately zero by the end. This change is economically meaningful. For example, a four-point deterioration in the negative unemployment gap ($x_t = -4$) implies an inflation response of about -0.44 percentage point at the 1982 slope, but almost no response at the 2012 slope.

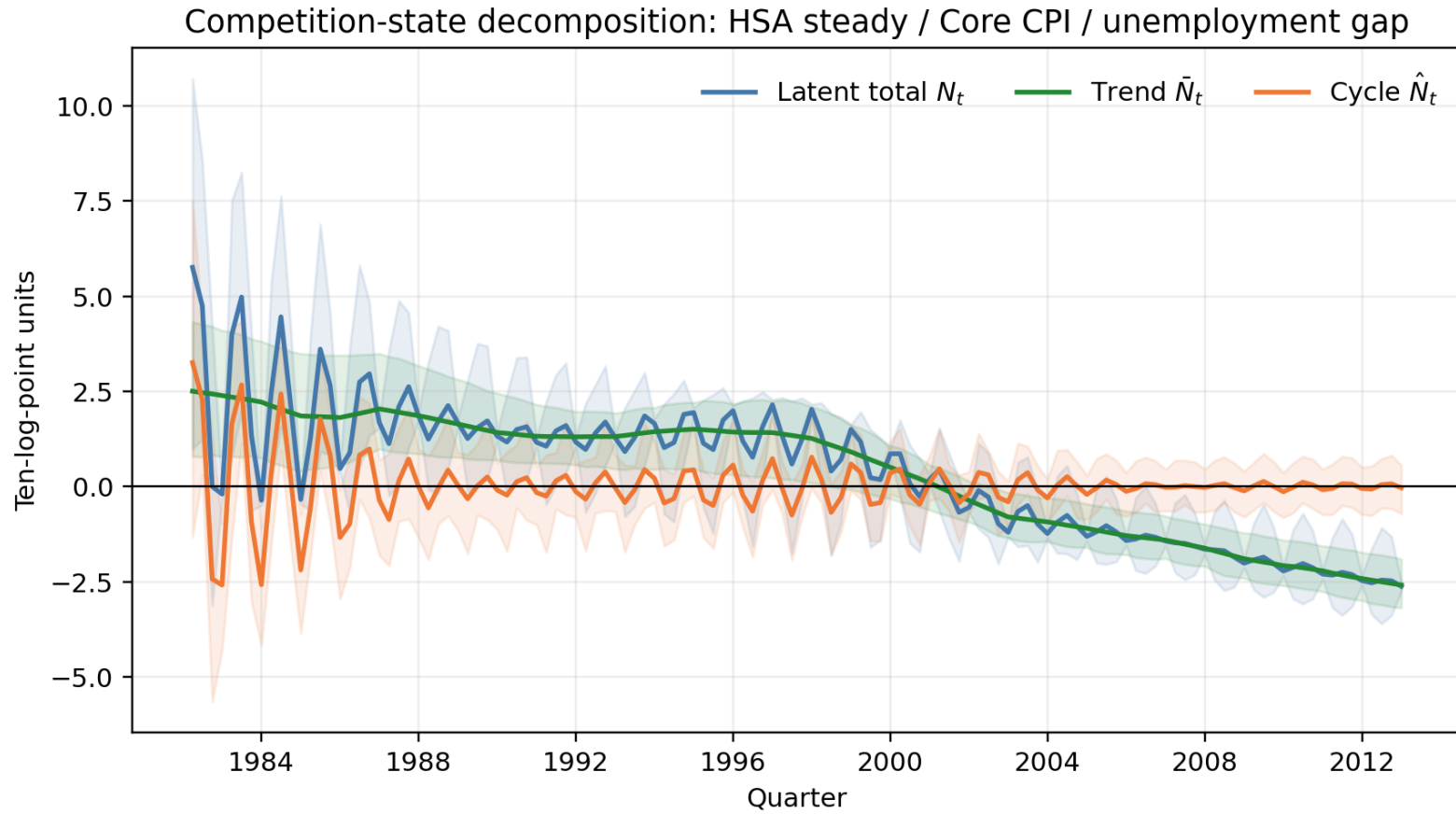


Figure 3: Posterior decomposition of the firm count $N_t = \bar{N}_t + \hat{N}_t$ under the main HSA steady (negative unemployment gap, core CPI), mixed-frequency design. Lines are posterior means, bands are 95% credible intervals. The quarterly states between the Q4 annual observations are inferred, not interpolated, so the bands widen between anchors.

5 The CES slope bias and the constant-slope benchmark

5.1 Theoretical prediction and its empirical counterpart

Fujiwara–Matsuyama (2026) show that because endogenous entry generates a cost-push term through strategic complementarity, a CES-type equation that regresses inflation on real marginal cost alone under-estimates the NKPC slope. In our HSA-dynamic representation, moving expected inflation to the left-hand side gives

$$y_t = \alpha a_t + \kappa_{\text{HSA}} x_t + q_t + e_t, \quad q_t = -\theta \hat{N}_t, \quad (30)$$

where $y_t = \pi_t - \mathbb{E}_t \pi_{t+1}$ and $a_t = \pi_{t-1} - \mathbb{E}_t \pi_{t+1}$. When the CES equation omits q_t ,

$$\text{Bias}(\hat{\kappa}_{\text{CES}}) = \text{plim}(\hat{\kappa}_{\text{CES}} - \hat{\kappa}_{\text{HSA}}) = \frac{\tilde{x}' \tilde{q}}{\tilde{x}' \tilde{x}}, \quad (31)$$

where tildes denote residuals after removing a_t and the activity shock $\zeta_t = x_t - \phi_1 x_{t-1}$. The theory predicts (31) is negative, i.e. $\kappa_{\text{CES}} - \kappa_{\text{HSA}} < 0$.

To keep this correspondence sharp, separately from the slack measures of the main results we use the BN-decomposed inverse markup as a proxy for real marginal cost and re-estimate CES and HSA dynamic with the same sample, priors, and MCMC settings. For each HSA posterior draw we form $q_t^{(s)} = -\theta^{(s)} \hat{N}_t^{(s)}$ and directly compute the bias in (31). Because the CES–HSA difference is a comparison across separately estimated posteriors, we build a Monte Carlo distribution of the difference by randomly permuting the HSA draws with a fixed seed.

5.2 Estimation results

For the inverse-markup specification that corresponds directly to the theory,

$$\begin{aligned} \kappa_{\text{CES}} &= +0.007[-0.217, +0.228], & \kappa_{\text{HSA}} &= +0.010[-0.212, +0.223], \\ \kappa_{\text{CES}} - \kappa_{\text{HSA}} &= -0.003[-0.306, +0.311], & \Pr(\kappa_{\text{CES}} - \kappa_{\text{HSA}} < 0) &= 0.513, \\ B_{\text{HSA}} &= +0.001[-0.014, +0.015], & \Pr(B_{\text{HSA}} < 0) &= 0.424 \end{aligned}$$

The CES–HSA difference does not show a negative bias. The omitted-variable bias implied by HSA itself also has a point estimate of the sign opposite to the theory (positive), with a credible interval that includes zero.

Table 5: CES–HSA-dynamic slope comparison and the omitted-variable bias implied by HSA. Each interval is a 95% credible interval. † marks that at least one of CES or HSA dynamic is outside the convergence criteria.

specification	CES kappa	HSA kappa	CES-HSA	Pr(CES-HSA<0)	HSA-implied OVB	Pr(OVB<0)	diagnostics
Inverse markup (theory)	+0.007 [-0.217, +0.228]†	+0.010 [-0.212, +0.223]†	-0.003 [-0.306, +0.311]†	0.513	+0.001 [-0.014, +0.015]†	0.424	watch†
Headline CPI / negative unemployment gap	+0.057 [-0.006, +0.120]†	+0.055 [-0.011, +0.117]†	+0.002 [-0.089, +0.091]†	0.482	+0.002 [-0.003, +0.023]†	0.393	watch†
Core CPI / negative unemployment gap	+0.040 [+0.007, +0.071]†	+0.040 [+0.009, +0.071]†	-0.000 [-0.044, +0.045]†	0.511	-0.000 [-0.002, +0.002]†	0.687	watch†
PPI / negative unemployment gap	+0.073 [-0.145, +0.284]†	+0.051 [-0.155, +0.263]†	+0.022 [-0.279, +0.318]†	0.441	+0.002 [-0.005, +0.012]†	0.310	watch†

Verdict. The inverse-markup row passes the convergence criteria for both models, but the slope intervals are wide and neither the CES–HSA difference nor the within-model OVB supports the negative sign the theory predicts. The three auxiliary rows using the negative unemployment gap give the same conclusion. This does not amount to a refutation of the theory: the inverse markup is an aggregate series rather than firm-level real marginal cost, and HSA dynamic is not a joint estimate of the full theoretical model. What we can say from this paper is only that *in the current data and semi-structural model, we do not obtain an estimate corresponding to the theory paper’s prediction of a negative bias in the CES slope.*

As a benchmark for the time-varying result of Section 4, we also estimate the constant-slope models. The full model-by-model coefficient tables for the negative unemployment gap, and the corresponding prior-sensitivity table, are collected in Appendix C.

CES vs HSA (constant κ) CES is the constant-slope benchmark. Among the converged cells, the core-CPI κ interval sits on the positive side, while headline CPI and PPI include zero. CES measures the existence of the average curve but does not explain the flattening. The prior-vs-posterior of the slope (Figure 4) shows CES’s constant κ and HSA steady’s slope at average competition κ_0 , both sharply updated by the data and sitting just above zero.

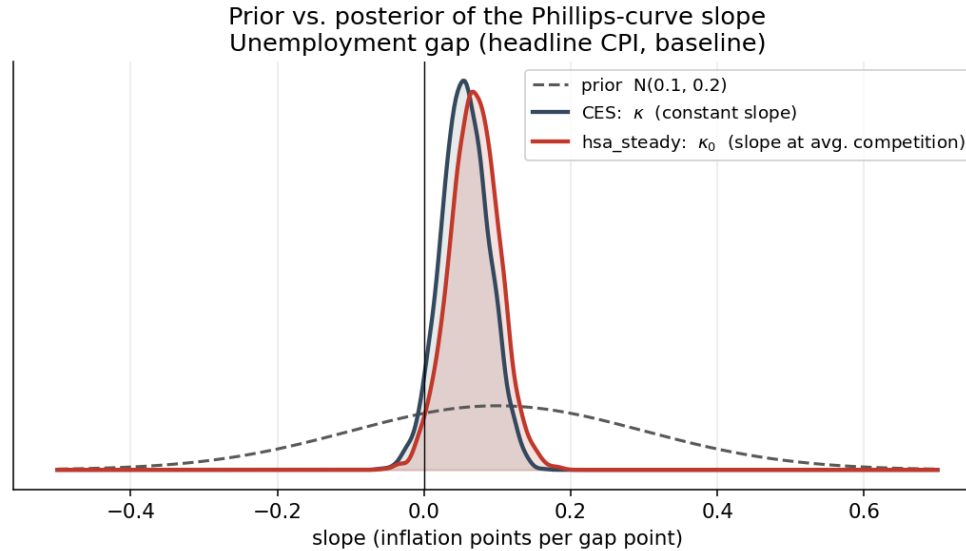


Figure 4: Prior vs. posterior of the Phillips-curve slope for the negative unemployment gap: CES’s constant slope κ and HSA steady’s slope at average competition κ_0 , against their common prior $\mathcal{N}(0.1, 0.2)$. Both are sharply updated by the data and sit just above zero; the two models agree on the slope level, so what distinguishes them is whether that slope varies with the firm count (Section 6).

6 Does the time-varying slope improve fit over the constant slope?

We compare CES (constant κ) against HSA steady (slope varying with the firm count) by two criteria: predictive fit (Table 6) and the Savage–Dickey Bayes factor for $\delta \neq 0$ (Table 7).

Fit, by four scores at once. Earlier revisions reported only an *in-sample plug-in* score. It evaluates the inflation equation at posterior-mean parameters and states (bars denote posterior means),

$$\hat{\pi}_t = \bar{\alpha} \pi_{t-1} + (1 - \bar{\alpha}) \mathbb{E}_t \pi_{t+1} + \bar{\kappa}_t x_t, \quad \bar{\kappa}_t = \bar{\kappa}_0 + \bar{\delta} \bar{N}_t \quad (\text{CES: } \bar{\kappa}_t = \bar{\kappa}), \quad (32)$$

and scores the residuals $r_t = \pi_t - \hat{\pi}_t$ by a Gaussian log-likelihood at the posterior-mean shock variance,

$$S = -\frac{T}{2} \log(2\pi\bar{\sigma}_e^2) - \frac{1}{2\bar{\sigma}_e^2} \sum_{t=1}^T r_t^2. \quad (33)$$

This reuses the estimation sample and collapses the posterior to a point, so it is neither an out-of-sample forecast nor a posterior predictive density.

We now report it *alongside* three proper scores rather than in place of them: the prequential one-step-ahead log predictive density $\text{LPD}_1 = \sum_t \log p(\pi_t \mid \pi_{1:t-1}, x, N)$, integrated over the posterior; WAIC; and PSIS-LOO. Keeping the plug-in score visible is the point of the table—it lets the reader see how much a crude criterion overstates the case:

Table 6: Fit comparison for the negative unemployment gap, mixed-frequency design. All four columns are log scores, so higher is better, and all are differences against nothing—only comparisons within a price index are meaningful. *Plug-in* is the in-sample score of the text: the inflation equation evaluated at posterior-mean parameters and states, scored by a Gaussian log-likelihood at the posterior-mean shock variance. LPD_1 is the prequential one-step-ahead log predictive density, $\sum_t \log p(\pi_t | \pi_{1:t-1}, x, N)$, integrated over the posterior. WAIC and PSIS-LOO are computed on the pointwise inflation log-likelihood. $\hat{k}$ is the maximum Pareto shape for the LOO importance weights; $\hat{k} > 0.7$ marks a cell where LOO is unreliable.

Price	Model	Plug-in	LPD_1	WAIC	PSIS-LOO	$\hat{k}$
Core CPI	CES	-18.66	-17.59	-22.98	-23.03*	0.84
	HSA steady	-10.73	-13.67	-19.69	-19.75	0.65
	HSA dynamic	-16.86	-15.14	-21.23	-21.27	0.54
	HSA full	-10.75	-12.22	-20.69	-20.73*	0.92
Headline CPI	CES	-119.94	-118.59	-125.25	-125.18	0.52
	HSA steady	-115.38	-115.84	-124.36	-124.35	0.61
	HSA dynamic	-119.80	-118.34	-125.45	-125.41	0.65
	HSA full	-115.94	-116.12	-124.21	-124.19	0.58
PPI	CES	-294.16	-290.56	-300.27	-300.17*	1.00
	HSA steady	-293.95	-290.16	-300.81	-301.09*	1.25
	HSA dynamic	-294.20	-291.62	-300.70	-300.74*	1.09
	HSA full	-293.95	-290.29	-299.76	-299.47	0.49

For HSA steady against CES the plug-in score gives +7.93 on core CPI where LPD_1 gives +3.92 and WAIC and LOO give +3.29 and +3.28; on headline CPI it gives +4.56 against +2.75, +0.89 and +0.83. The plug-in score overstates the improvement by a factor of roughly two to five, which is what one expects from a criterion that rewards a model for the extra latent states it fitted to this same sample. The *ordering* nevertheless survives: on both CPI indices the time-varying slope improves every score, on PPI no score separates the models (LPD_1 +0.40, WAIC −0.54, LOO −0.92), and HSA dynamic—the entry channel without a time-varying slope—never beats HSA steady. Two caveats. First, the LOO column is unreliable wherever the maximum Pareto $\hat{k}$ exceeds 0.7, which is 5 of the 12 cells: CES and HSA full on core CPI (0.84 and 0.92), and CES, HSA steady and HSA dynamic on PPI (1.00, 1.25, 1.09). That includes the CES baseline for the core-CPI comparison, so the +3.28 LOO difference quoted above rests on one number that PSIS itself flags; the WAIC difference of +3.29, which does not depend on importance sampling, is the more trustworthy of the two, and it agrees. The PPI cells should simply not be read off the LOO column. Second, all four scores are computed on the estimation sample— LPD_1 is prequential in the sense that each π_t is scored against a predictive density formed without π_t , but the parameters were estimated on the full sample—so none of this is a genuine out-of-sample exercise.

Bayes factor (Savage–Dickey), stated precisely. We test $\delta = 0$ *within the HSA state model*: M_1 is HSA steady with δ free and M_0 is the same model with $\delta = 0$. The Savage–Dickey identity gives

$$\text{BF}_{10} = \frac{p(\delta = 0)}{\pi(\delta = 0 \mid y)}, \quad p(\delta = 0) = \frac{1}{\sigma_\delta \sqrt{2\pi}} = \frac{1}{0.02\sqrt{2\pi}} \approx 19.95 \quad (\delta \sim \mathcal{N}(0, \sigma_\delta^2)), \quad (34)$$

The posterior ordinate is computed *without density estimation*. The Gibbs update for the inflation coefficients is an exact Gaussian conditional, so the marginal ordinate is its Rao–Blackwellised average over the retained draws of everything else, $\psi = (\bar{N}_{0:T}, \hat{N}_{0:T}, \sigma_\eta^2, \lambda_{e\zeta}, \phi_1)$:

$$\hat{\pi}(\delta = 0 \mid y) = \frac{1}{S} \sum_{s=1}^S \mathcal{N}(0; b_1^{(s)}[\delta], V_1^{(s)}[\delta, \delta]), \quad \psi^{(s)} \sim p(\psi \mid y), \quad (35)$$

with $b_1^{(s)}, V_1^{(s)}$ the conditional moments of (20) evaluated at $\psi^{(s)}$. This matters because $\delta = 0$ lies about three posterior standard deviations outside the sampled range—no retained draw of δ is negative—so a kernel density evaluated there extrapolates its own tail rather than measuring the posterior. For the main cell the kernel estimate moves from 22 to 6.6×10^4 as its bandwidth is scaled between $3\times$ and $0.5\times$ Scott’s rule, whereas (35) draws on 1,504 effectively contributing terms out of 3,200 (the three largest carry 1.2% of the sum) and has a Monte Carlo standard error of about 3%.

Three cautions remain. (i) This is *not* a CES-vs-HSA marginal-likelihood ratio: CES and HSA also differ in the latent- N block, which the SDDR holds fixed, so we keep the comparison inside the HSA state model. (ii) The theory is one-sided ($\delta > 0$); the two-sided BF_{10} is conservative. (iii) A Bayes factor against a point null is proportional to the prior width by construction, so these numbers are statements about $\delta = 0$ under the stated $\mathcal{N}(0, 0.02^2)$ prior and are not prior-free.

Table 7: Constant vs. time-varying slope: is $\delta \neq 0$? Posterior mean of δ [95% CI] with the Savage–Dickey Bayes factor BF_{10} for $\delta \neq 0$ within the HSA state model in parentheses (HSA steady, baseline priors, quarterly, unrestricted). Values are auto-generated from the current posteriors; $\text{BF}_{10} > 1$ favors a concentration-dependent slope.

Activity measure	Headline CPI	Core CPI	PPI
Negative unemployment gap	+0.029 [+0.005, +0.053] (11.2)	+0.022 [+0.008, +0.038] (36.2)	+0.009 [-0.028, +0.047] (1.1)
Output gap (HP)	-0.006 [-0.039, +0.027] (0.9)	+0.028 [+0.009, +0.047] (19.6)	-0.004 [-0.043, +0.035] (1.0)
Output gap (BN)	-0.019 [-0.052, +0.013] (1.6)	+0.000 [-0.031, +0.025] (0.7)	-0.008 [-0.046, +0.031] (1.0)

Verdict. For the negative unemployment gap, allowing the slope to vary with the firm count improves the in-sample fit score and the Bayes factor favors $\delta \neq 0$ strongly under both CPI indices. For the headline output gaps the gain is marginal, consistent with the appendix tables: a positive competition dependence appears only in the core-CPI HP specification, not in BN. For the labor-share gap and inverse markup there is no inflation signal, so constant and time-varying slopes fit equally well. We stress that this is an in-sample fit comparison, not evidence of out-of-sample predictive gains.

Limits of the model comparison. We read the Savage–Dickey ratios only as the point null $\delta = 0$ *within the HSA state model*; they are not a CES-vs-HSA marginal-likelihood ratio. The in-sample LPPD and RMSE conditioned on the latent states are likewise not out-of-sample predictions, so they are not used as grounds to select a more complex model. A corrected conditional marginal likelihood $p(\pi \mid x, N^{\text{obs}}, M)$ —states integrated out exactly by the Kalman filter, Chib ordinate with explicit reduced Gibbs runs—now exists in the repository (Section 11), but its numbers are not promoted into this table until they have been reviewed.

7 What PCHIP interpolation does to the estimates

The conventional alternative to the mixed-frequency design of Section 3.3 is to interpolate the annual firm count to a quarterly series and then use that series *as data*. This section reports that version and documents what the interpolation costs. Everything else—3 price indices, 3 activity measures, 5 models, 3 priors, the inverse-markup slope-bias diagnostic, and all MCMC settings—is identical. CES contains neither N_t nor a latent firm-count state, so its likelihood and posterior are invariant to the observation scheme and its 16 cells are shared across both designs.

7.1 The two designs, side by side

Setting $\mathcal{T}_N = \{1, \dots, T\}$ in (12)—every quarter observed—recovers the interpolated filter exactly; the mixed-frequency design is that same filter with 93 of the 124 firm-count rows removed. The estimator is not the issue. What differs is what is fed to it.

Table 8: What the interpolation changes. Posterior quantities for the main cell (HSA steady, core CPI, negative unemployment gap, baseline priors).

	Mixed-frequency (main)	PCHIP-interpolated
Firm-count observations used	31 of 124	124 of 124
Estimated measurement error σ_N	0.068	0.023
ρ_1, ρ_2	+0.20, −0.89	+1.80, −0.81
Implied AR(2) period	4.3 quarters	82 quarters
Mean reversion $1 - \rho_1 - \rho_2$	1.69	0.014
Posterior corr($\tilde{N}_0, \hat{N}_0$)	+0.13	− 0.9996
Posterior corr($\tilde{N}$ level, κ_0)	−0.32	−0.55
δ	+0.022 [+0.008, +0.038]	+0.023 [+0.010, +0.039]
κ_t start $\rightarrow$ end	+0.110 $\rightarrow$ −0.002	+0.120 $\rightarrow$ −0.005

Three things follow. First, the interpolated series is so smooth that the model drives σ_N down to 0.023 to fit it, which pins $\tilde{N}_t + \hat{N}_t$ at all 124 quarters and leaves only the split free. Second, the AR(2) cycle is pushed to a near-unit root, $\rho_1 + \rho_2 = 0.986$, so its mean-reversion term $1 - \rho_1 - \rho_2$ collapses to 0.014. That term is the *only* thing anchoring the level of $\hat{N}_t$: adding a constant c to $\tilde{N}$ and subtracting it from $\hat{N}$ leaves the firm-count equation, the trend equation and—through $\kappa_0 \rightarrow \kappa_0 - \delta c$ —the inflation equation all exactly unchanged. When the anchor vanishes the two states become almost perfectly counter-identified, which is the −0.9996 in the table. Across all 122 estimated HSA cells the correlation between $1 - \rho_1 - \rho_2$ and $\text{corr}(\tilde{N}_0, \hat{N}_0)$ is +0.92, so this is the mechanism and not a coincidence. Third, the “cycle” the interpolated design recovers has an 82-quarter period: it is a second trend competing with $\tilde{N}_t$ for the same low-frequency variation, not a business-cycle object.

What is *not* affected. δ is essentially unchanged—the interaction $x_t\bar{N}_t$ that identifies it is insensitive to a level shift in $\bar{N}$, which is absorbed by κ_0 . Across the nine HSA-steady baseline cells the two designs agree on δ to the third decimal in eight of them. The substantive conclusion of this paper does not depend on the choice; the reported precision and the interpretation of the latent states do.

7.2 Interpolated results, for comparison

Table 9: Headline results (main model: HSA steady, baseline priors, PCHIP-interpolated, $T = 124$), by price index and activity measure. δ is the competition dependence of the slope (a positive δ means the slope flattens as the firm count falls); BF_{10} is the Savage–Dickey Bayes factor against $\delta = 0$; κ_0 is the slope at average competition; the last column is the implied κ_t from the start to the end of the sample. The **main specification (core CPI, negative unemployment gap)** is in bold. †: fails the *coefficient* convergence rule ($\hat{R} \leq 1.01$ and bulk ESS ≥ 400 over every scalar parameter, including the trend drift n and all variances). “OK (coef)” marks a cell that passes on the coefficients but not on the latent-state paths; the group-by-group diagnostics are in Table 13. The $\delta \hat{R}/\text{ESS}$ column is that coefficient’s own mixing, which the blanket flag does not show.

Activity measure	Price	δ [95% CI]	BF_{10}	κ_0 [95% CI]	κ_t start→end	$\delta \hat{R}/\text{ESS}$	Conver
Negative unemployment gap	Core CPI	+0.023 [+0.010, +0.039]	106.8	+0.051 [+0.016, +0.090]	+0.120 → -0.005	1.001 / 1538	OK (coef)
	Headline CPI	+0.030 [+0.008, +0.051]	22.4	+0.067 [+0.002, +0.136]	+0.163 → -0.010	1.000 / 2912	OK (coef)
	PPI	+0.012 [-0.024, +0.049]	1.1	+0.075 [-0.131, +0.279]	+0.113 → +0.044	1.000 / 3004	OK (coef)
HP output gap	Core CPI	+0.029 [+0.011, +0.048]	67.7	+0.069 [+0.018, +0.123]	+0.161 → -0.005	1.000 / 2587	OK (coef)
	Headline CPI	-0.002 [-0.031, +0.028]	0.7	+0.141 [+0.048, +0.237]	+0.135 → +0.146	1.000 / 3144	OK (coef)
	PPI	-0.005 [-0.044, +0.034]	1.0	+0.210 [-0.067, +0.473]	+0.193 → +0.224	1.000 / 3121	OK (coef)
BN output gap	Core CPI	+0.011 [-0.008, +0.031]	0.9	+0.031 [-0.020, +0.083]	+0.063 → +0.005	1.000 / 2796	OK (coef)
	Headline CPI	-0.013 [-0.042, +0.016]	1.1	+0.064 [-0.024, +0.153]	+0.022 → +0.097	1.002 / 2729	OK (coef)
	PPI	-0.007 [-0.044, +0.030]†	1.1	+0.190 [-0.054, +0.427]	+0.167 → +0.210	1.000 / 3177	wat

Table 10: Interpolated design: model-by-model results for the negative unemployment gap.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.057 [-0.006, +0.120]	—	—	—	—	—	OK
CES	Core CPI	+0.040 [+0.007, +0.071]	—	—	—	—	—	OK
CES	PPI	+0.073 [-0.145, +0.284]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.067 [+0.002, +0.136]	+0.030 [+0.008, +0.051]	22.4	—	—	+0.163 → -0.010	OK
HSA steady	Core CPI	+0.051 [+0.016, +0.090]	+0.023 [+0.010, +0.039]	106.8	—	—	+0.120 → -0.005	OK (coef)
HSA steady	PPI	+0.075 [-0.131, +0.279]	+0.012 [-0.024, +0.049]	1.1	—	—	+0.113 → +0.044	OK
HSA dynamic	Headline CPI	+0.052 [-0.026, +0.143]	—	—	-0.013 [-0.144, +0.166]	—	—	watch
HSA dynamic	Core CPI	+0.032 [-0.004, +0.066]	—	—	-0.022 [-0.076, +0.039]	—	—	OK
HSA dynamic	PPI	+0.057 [-0.153, +0.276]	—	—	+0.011 [-0.296, +0.343]	—	—	watch
HSA const-theta	Headline CPI	+0.080 [+0.007, +0.156]	+0.033 [+0.007, +0.059]	14.4	+0.036 [-0.083, +0.161]	—	+0.183 → -0.002	OK
HSA const-theta	Core CPI	+0.066 [+0.021, +0.106]	+0.027 [+0.012, +0.042]	202.7	+0.040 [-0.027, +0.106]	—	+0.156 → -0.008	OK
HSA const-theta	PPI	+0.078 [-0.135, +0.300]	+0.011 [-0.027, +0.050]	1.1	+0.020 [-0.283, +0.349]	—	+0.111 → +0.051	watch
HSA full	Headline CPI	+0.088 [+0.011, +0.161]	+0.033 [+0.006, +0.061]	10.3	+0.070 [-0.057, +0.195]	-0.015 [-0.052, +0.023]	+0.160 → +0.018	watch
HSA full	Core CPI	+0.062 [+0.020, +0.105]	+0.025 [+0.007, +0.043]	27.5	+0.052 [-0.027, +0.132]	-0.010 [-0.043, +0.021]	+0.137 → -0.005	watch
HSA full	PPI	+0.080 [-0.145, +0.301]	+0.011 [-0.028, +0.049]	1.1	+0.043 [-0.262, +0.373]	-0.005 [-0.043, +0.035]	+0.111 → +0.054	watch

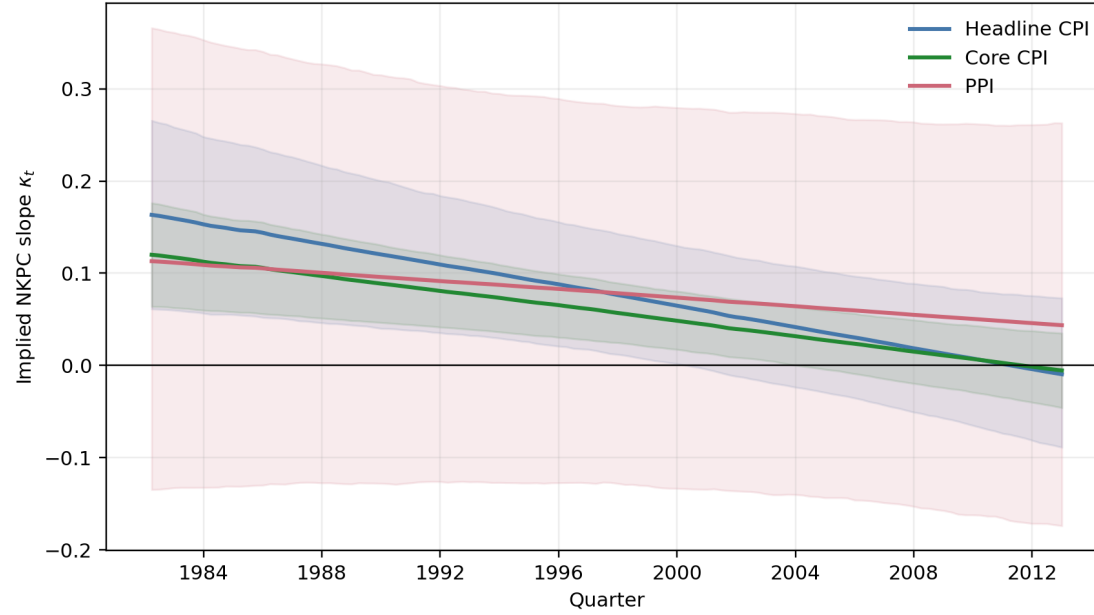


Figure 5: Interpolated design: time-varying slope κ_t for the negative unemployment gap. Compare Figure 2: the paths are close, but the bands here do not include the uncertainty from the 93 firm-count values that were interpolated rather than observed.

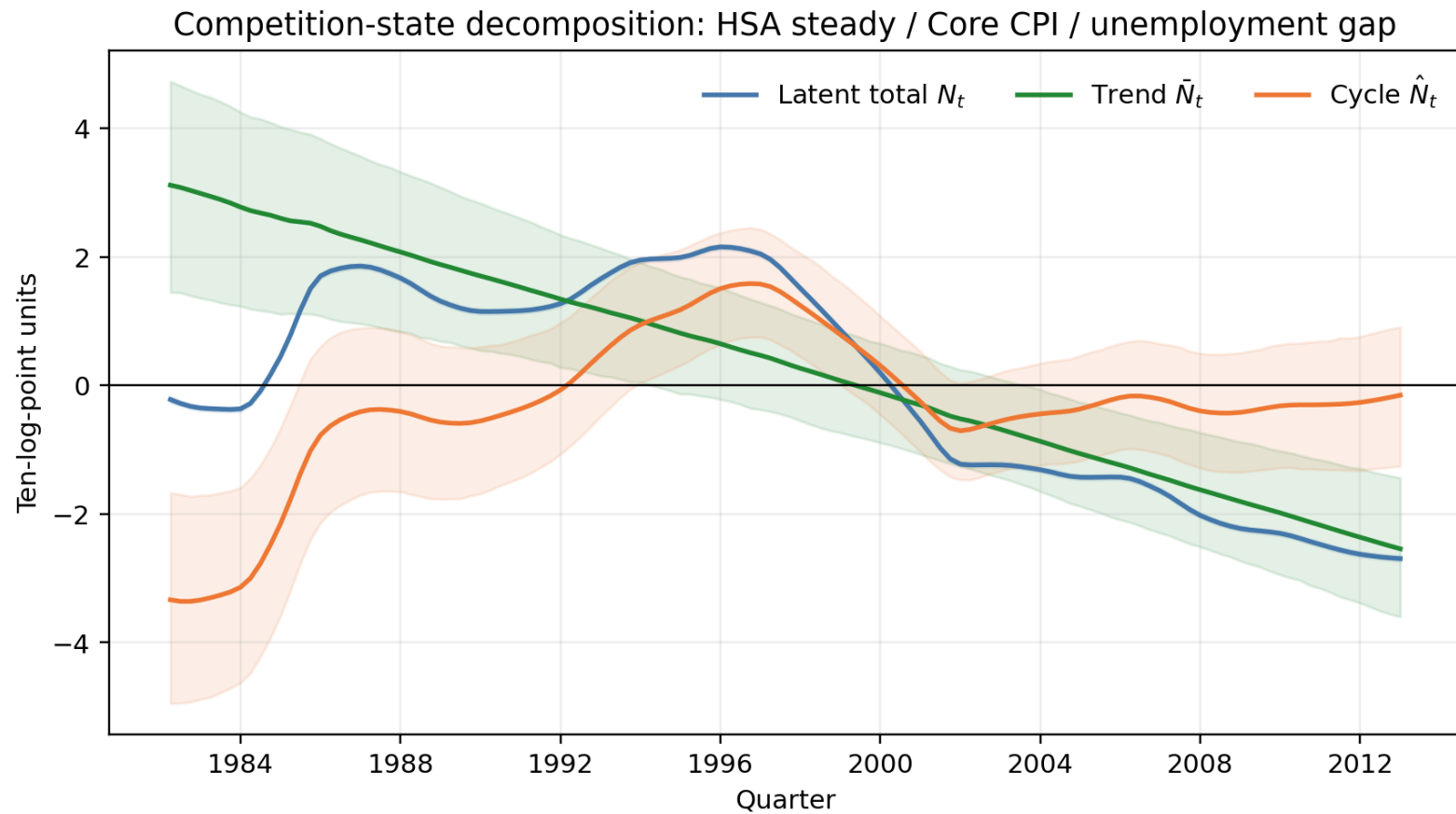


Figure 6: Interpolated design: firm-count state decomposition. The cycle $\hat{N}_t$ reaches ± 3.4 ten-log-point units—a ± 34 log-point “cyclical” swing in the number of listed firms—which is the level ridge, not an economic cycle. Compare Figure 3.

7.3 Prior sensitivity under interpolation

Table 11: Interpolated design: three-prior sensitivity for the negative unemployment gap. † marks cells outside the convergence criteria.

model	inflation	parameter	baseline	weak	tight
CES	Core CPI	κ	+0.040 [+0.007, +0.071]	+0.040 [+0.010, +0.071]	+0.039 [+0.002, +0.076]
CES	Headline CPI	κ	+0.057 [-0.006, +0.120]	+0.056 [-0.011, +0.121]	+0.059 [-0.006, +0.124]
CES	PPI	κ	+0.073 [-0.145, +0.284]	+0.060 [-0.181, +0.303]	+0.088 [-0.061, +0.241]
HSA const-theta	Core CPI	δ	+0.027 [+0.012, +0.042]	+0.035 [+0.017, +0.071] [†]	+0.014 [-0.002, +0.028] [†]
HSA const-theta	Headline CPI	δ	+0.033 [+0.007, +0.059]	+0.065 [+0.024, +0.122] [†]	+0.015 [-0.003, +0.031]
HSA const-theta	PPI	δ	+0.011 [-0.027, +0.050] [†]	+0.073 [-0.079, +0.204] [†]	+0.004 [-0.016, +0.023]
HSA dynamic	Core CPI	κ	+0.032 [-0.004, +0.066]	+0.034 [-0.006, +0.071] [†]	+0.031 [-0.009, +0.071]
HSA dynamic	Headline CPI	κ	+0.052 [-0.026, +0.143] [†]	+0.075 [-0.012, +0.161] [†]	+0.043 [-0.027, +0.109]
HSA dynamic	PPI	κ	+0.057 [-0.153, +0.276] [†]	+0.045 [-0.247, +0.348] [†]	+0.084 [-0.065, +0.245]
HSA full	Core CPI	δ	+0.025 [+0.007, +0.043] [†]	+0.047 [+0.004, +0.134] [†]	+0.016 [+0.002, +0.029] [†]
HSA full	Headline CPI	δ	+0.033 [+0.006, +0.061] [†]	+0.044 [-0.010, +0.119] [†]	+0.014 [-0.003, +0.030] [†]
HSA full	PPI	δ	+0.011 [-0.028, +0.049] [†]	+0.073 [-0.063, +0.207] [†]	+0.004 [-0.015, +0.023] [†]
HSA steady	Core CPI	δ	+0.023 [+0.010, +0.039]	+0.037 [+0.013, +0.109] [†]	+0.013 [+0.003, +0.024]
HSA steady	Headline CPI	δ	+0.030 [+0.008, +0.051]	+0.056 [+0.018, +0.133] [†]	+0.016 [+0.000, +0.030]
HSA steady	PPI	δ	+0.012 [-0.024, +0.049]	+0.073 [-0.061, +0.189] [†]	+0.004 [-0.015, +0.024]

Table 12: Interpolated design: model-by-model convergence tally.

model	sampler	runs	warnings	state warnings	joint warnings	max Rhat	min bulk ESS
CES	conjugate Gibbs	16	0	0	0	1.002	2622
HSA const-theta	joint FFBS	15	9	7	9	1.301	5
HSA dynamic	joint FFBS	16	10	9	10	1.021	60
HSA full	Particle Gibbs	15	15	15	15	1.968	3
HSA steady	joint FFBS	15	4	3	5	1.019	229

Table 13: Interpolated design: scalar / state-path / derived-path diagnostics by model. Compare Table 21: the worst-mixing scalar here is the trend drift n in nearly every cell, which is the level ridge in the parameter block.

model	inflation	sampler	scalar Rhat	scalar ESS	worst scalar	n Rhat	n ESS	state Rhat	state ESS	derived Rhat	derived ESS	joint
CES	Headline CPI	conjugate Gibbs	1.001	3049	phi_1	—	—	—	—	—	—	OK
CES	Core CPI	conjugate Gibbs	1.001	2905	phi_1	—	—	—	—	—	—	OK
CES	PPI	conjugate Gibbs	1.002	3023	sigma_e	—	—	—	—	—	—	OK
HSA steady	Headline CPI	joint FFBS	1.001	732	n	1.001	732	1.003	912	1.002	2524	OK
HSA steady	Core CPI	joint FFBS	1.008	431	n	1.008	431	1.011	431	1.000	3069	watch
HSA steady	PPI	joint FFBS	1.004	488	n	1.004	488	1.003	526	1.001	2966	OK
HSA dynamic	Headline CPI	joint FFBS	1.001	138	n	1.000	138	1.002	147	—	—	watch
HSA dynamic	Core CPI	joint FFBS	1.004	542	n	1.004	542	1.003	616	—	—	OK
HSA dynamic	PPI	joint FFBS	1.002	361	n	1.001	361	1.002	395	—	—	watch
HSA const-theta	Headline CPI	joint FFBS	1.001	498	n	1.001	498	1.002	618	1.001	1703	OK
HSA const-theta	Core CPI	joint FFBS	1.003	689	n	1.003	689	1.003	797	1.001	1926	OK
HSA const-theta	PPI	joint FFBS	1.007	319	n	1.007	319	1.007	349	1.002	872	watch
HSA full	Headline CPI	Particle Gibbs	1.176	8	n	1.176	8	1.560	4	1.012	739	watch
HSA full	Core CPI	Particle Gibbs	1.214	7	n	1.214	7	1.433	4	1.018	156	watch
HSA full	PPI	Particle Gibbs	1.572	3	n	1.572	3	1.944	3	1.101	13	watch

A prior–data conflict that is specific to interpolation. The AR(2) prior, $\rho_1 \sim \mathcal{N}(0.5, \sigma_\rho^2)$ and $\rho_2 \sim \mathcal{N}(-0.5, \sigma_\rho^2)$ truncated to the stationary region, encodes a cycle of about 5.2 quarters with a two-quarter half-life, and does so identically under all three prior sets—they differ only in width. Under the mixed-frequency design the posterior is compatible with it: the ρ_1 posterior sits between -1.5 and $+1.6$ prior standard deviations of the prior mean. Under interpolation it sits $+2.7$ to $+7.1$ prior standard deviations away. The prior is not obviously wrong; the interpolated series is simply not the object the prior describes. One consequence is that the tight-prior cells inherit a prior–data tug-of-war in the AR(2) block, which is where their mixing failures come from.

8 Scope condition and the limits of identification

For honesty we state where the main result does *not* hold and what it can and cannot claim.

- **Not identified for PPI.** Switching to PPI, closer to the theory’s left-hand side, makes δ include zero for all activity measures. Despite the measurement limitation that no PPI expectation exists, we cannot claim that the CPI relationship holds generally at the price-setting stage.
- **HSA full does not converge; const-theta now does, in part.** Every HSA-full cell is outside the acceptance criteria even under Particle Gibbs, so γ —which only HSA full estimates—remains undemonstrated. HSA const-theta is a different case: its earlier failure was a blocking artifact of the alternating FFBS, and under the exact joint FFBS 6 of its 15 cells now pass, including both CPI negative-unemployment-gap cells. Its δ agrees in sign and magnitude with HSA steady there. That is a computational fix, not new identification.
- **The joint posterior does not converge, though δ does.** Every HSA cell in the main design is outside the whole-run acceptance criteria. The failure is concentrated in the AR(2) block: (ρ_1, ρ_2) is the worst-mixing scalar in 53 of the 61 HSA cells, and the latent-state paths fail alongside it. What does *not* fail is the coefficient the conclusions rest on: δ has $\hat{R} = 1.000$ and bulk ESS above 2,600 in eight of the nine HSA-steady baseline cells, and δ and κ_0 fail at exactly the same rate under either observation design (8 and 6 cells out of 61). We therefore report δ and κ_t as results and treat the estimated cycle $\hat{N}_t$, its AR(2) coefficients, and anything that depends on them as diagnostic only. Readers who want the whole-run criterion satisfied should treat every number in this paper as provisional.

The latent-level ridge, and where it comes from. The firm-count equation $N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t$ constrains the *sum* of the two states but not the split. Adding a constant c to $\bar{N}$ and subtracting it from $\hat{N}$ leaves that equation, the random-walk trend equation, and—through $\kappa_0 \rightarrow \kappa_0 - \delta c$ —the inflation equation all exactly unchanged. The only thing that resists is the AR(2) cycle equation, which carries no intercept and therefore pulls $\hat{N}$ toward mean zero with force $1 - \rho_1 - \rho_2$; the initial-state prior $s_0 \sim \mathcal{N}(0, 10I_3)$ resists weakly as well.

Earlier revisions of this paper described the resulting ridge as a limitation of the data. It is better described as a consequence of the interpolated design. Under interpolation the AR(2) is pushed to a near-unit root and $1 - \rho_1 - \rho_2$ collapses to 0.014, so the anchor effectively disappears and the posterior correlation between the two states reaches -0.9996 . Under the mixed-frequency design used here the same quantity is 1.69 and the correlation is $+0.13$. Across all 122 estimated HSA cells the two quantities correlate at $+0.92$, which identifies the mechanism (Section 7).

Two things follow. First, κ_0 —the slope at average competition—is identified only up to the location normalisation of $\bar{N}$, and under interpolation that normalisation comes substantially from the initial-state prior rather than from data: the posterior correlation between the $\bar{N}$ level and κ_0 is -0.55 there against -0.32 here. Statements about the *level* of the Phillips-curve slope should be read with that in mind. Second, δ is not affected, because a level shift in $\bar{N}$ is absorbed by κ_0 and leaves the interaction $x_t \bar{N}_t$ that identifies δ unchanged. That is why the headline result is stable across designs while the state decomposition is not.

Not a causal effect. $\bar{N}_t$ is a near-monotone low-frequency trend, so on this design δ cannot be separated from other things that trended over the same decades: the Volcker monetary-policy regime change, the anchoring of inflation expectations, changes in labor-market institutions and unionization, globalization and import competition, changes in the frequency of price adjustment, or a simple time trend / structural break.

The most we can say is that *the low-frequency decline in the firm count and the decline in inflation's sensitivity to the negative unemployment gap occurred together, with a sign consistent with the HSA theory.* We therefore avoid “competition causes flattening” and “we identify the mechanism.”

9 Robustness

We judge robustness on three points: (i) how the sign, interval, and magnitude change under alternative priors, (ii) whether the result survives switching the price index to PPI, and (iii) whether the same conclusion holds using only converged runs. The prior check re-estimates *every cell* under baseline, weak, and tight priors (Table 14); it is a genuine re-estimation of the current runs, not a re-weighting of a single posterior.

Table 14: Three-prior sensitivity under the main mixed-frequency design (negative-unemployment-gap specification): δ under baseline, weak, and tight priors, by model and price index. Each cell is a separate re-estimation, not a re-weighting. The interpolated counterpart is Table 11.

model	inflation	parameter	baseline	weak	tight
CES	Core CPI	κ	+0.040 [+0.007, +0.071]	+0.040 [+0.010, +0.071]	+0.039 [+0.002, +0.076]
CES	Headline CPI	κ	+0.057 [-0.006, +0.120]	+0.056 [-0.011, +0.121]	+0.059 [-0.006, +0.124]
CES	PPI	κ	+0.073 [-0.145, +0.284]	+0.060 [-0.181, +0.303]	+0.088 [-0.061, +0.241]
HSA const-theta	Core CPI	δ	+0.022 [+0.004, +0.038] [†]	+0.034 [+0.016, +0.053] [†]	+0.008 [-0.010, +0.024] [†]
HSA const-theta	Headline CPI	δ	+0.029 [+0.003, +0.055] [†]	+0.053 [+0.015, +0.109] [†]	+0.010 [-0.009, +0.027] [†]
HSA const-theta	PPI	δ	+0.009 [-0.030, +0.047] [†]	+0.069 [-0.056, +0.193] [†]	+0.002 [-0.017, +0.022] [†]
HSA dynamic	Core CPI	κ	+0.040 [+0.009, +0.071] [†]	+0.033 [-0.007, +0.070] [†]	+0.040 [+0.005, +0.076] [†]
HSA dynamic	Headline CPI	κ	+0.055 [-0.011, +0.117] [†]	+0.047 [-0.036, +0.130] [†]	+0.060 [-0.002, +0.123]
HSA dynamic	PPI	κ	+0.051 [-0.155, +0.263] [†]	+0.020 [-0.249, +0.268] [†]	+0.083 [-0.072, +0.236] [†]
HSA full	Core CPI	δ	+0.025 [+0.008, +0.043] [†]	+0.029 [+0.003, +0.058] [†]	+0.007 [-0.009, +0.024] [†]
HSA full	Headline CPI	δ	+0.029 [+0.002, +0.054] [†]	+0.049 [+0.001, +0.098] [†]	+0.010 [-0.008, +0.027] [†]
HSA full	PPI	δ	+0.010 [-0.029, +0.047] [†]	+0.076 [-0.040, +0.188] [†]	+0.002 [-0.018, +0.022] [†]
HSA steady	Core CPI	δ	+0.022 [+0.008, +0.038] [†]	+0.038 [+0.013, +0.124] [†]	+0.009 [-0.007, +0.024] [†]
HSA steady	Headline CPI	δ	+0.029 [+0.005, +0.053] [†]	+0.052 [+0.017, +0.116] [†]	+0.010 [-0.008, +0.027] [†]
HSA steady	PPI	δ	+0.009 [-0.028, +0.047] [†]	+0.075 [-0.039, +0.188] [†]	+0.002 [-0.017, +0.021] [†]

HSA steady. The headline-CPI and core-CPI δ keep a positive sign under weak and baseline priors, but not their interval: core CPI moves +0.038 [+0.013, +0.124] (weak), +0.022 [+0.008, +0.038] (baseline), +0.009 [-0.007, +0.024] (tight), so the tight prior pulls the interval across zero. The result is *not* prior-invariant, and the paper does not claim otherwise. PPI includes zero under all three priors.

Because the three prior sets move every prior at once, we ran a factorial experiment crossing δ 's own prior with the AR(2) prior while holding all others at baseline, to see which channel drives the sweep. The answer is δ 's own prior: tightening it alone accounts for 84% of the headline-CPI shrinkage and 78% of the core-CPI shrinkage, against 13% and 50% for the AR(2) prior (with a -28% interaction in the core cell). This matters because the AR(2) prior is in tension with the interpolated data (Section 7); the experiment shows that tension is not what is moving δ . It does, however, break the sampler in the tight corner: the cells where the AR(2) prior fights the data hardest are where δ 's own ESS collapses.

Model variants. HSA const-theta reproduces the main result almost exactly under the added entry term: $\delta = +0.029 [+0.003, +0.055]$ for headline CPI and $\delta = +0.022 [+0.004, +0.038]$ for core CPI, against $+0.029$ and $+0.022$ under HSA steady. The agreement is close enough that the firm-count-trend channel is evidently not being carried by the omission of the entry term. Under the interpolated design the same comparison also holds, and there 6 of the 15 const-theta cells additionally satisfy the coefficient rule; under mixed frequency none do, for the AR(2) reason above. HSA full remains outside the criteria in every cell under either design, so γ is not identified here. HSA dynamic is short of ESS in most cells once the trend drift n and the variance block are included in the rule.

Period robustness. The state equation assumes consecutive quarters. We do not use the old treatment that deletes an interior span and compresses what precedes and follows. The current 77 runs are full-sample comparisons; until a separate contiguous subsample estimation of the corrected revision is done, we do not include pre-2008 or post-1988 numbers in the conclusions.

10 Conclusion

Over 1982–2012 in the United States, the low-frequency decline in the number of listed firms and the decline in inflation’s sensitivity to the negative unemployment gap moved together. We read this as theory-consistent time-series evidence, not a causal effect.

- **Sign consistent with HSA.** For the core-CPI negative-unemployment-gap specification $\delta = +0.022[+0.008, +0.038]$ ($\text{BF}_{10} = 36.2$); a falling firm count implies a flatter slope, as the theory predicts.
- **Economically sizable flattening.** The implied κ_t falls from $+0.110$ to -0.002 ; the start-to-end response to a one-point negative unemployment gap differs by about 0.11 point.
- **Core CPI aligns the output gap.** Under core CPI the HP output gap shows the same positive sign; BN stays inconclusive.
- **Not confirmed for PPI.** The negative-unemployment-gap specification gives $\delta = +0.009[-0.028, +0.047]$, with the interval including zero for all activity measures, so the relationship is not a general upstream (price-setting-stage) effect.
- **No CES-slope bias found.** The theory’s predicted negative bias of the CES slope is not supported: both the CES–HSA difference and the HSA-implied omitted-variable bias have intervals including zero, with, if anything, positive point estimates.
- **Cyclical-entry channel unconfirmed, and sign-unstable.** The theory predicts a positive entry coefficient θ in $-\theta_t \hat{N}_t$. In the main design HSA const-theta gives $+0.022[-0.159, +0.241]$ for headline CPI and $-0.039[-0.136, +0.054]$ for core CPI—opposite signs across two price indices, both intervals wide and straddling zero. The sign is not stable across the observation design either: the same core-CPI cell gives $+0.040$ with $\Pr(\theta > 0) = 0.89$ under interpolation against -0.039 with $\Pr(\theta > 0) = 0.15$ here. Under interpolation the sign also flips between activity measures ($+0.040$ core CPI with the unemployment gap versus -0.049 with the BN output gap, the latter putting 97% of its mass on the theory-wrong side). We claim no support for this channel and note that what the data say about its sign depends on choices that should not matter. γ is estimated only in HSA full, which converges in no cell under either design.
- **The result does not depend on the interpolation, but the reported precision does.** Estimating with the annual firm count PCHIP-interpolated to quarterly and used as data gives $\delta = +0.023[+0.010, +0.039]$ against $+0.022[+0.008, +0.038]$ here—the same number. What the interpolation changes is everything around it: it drives the firm-count measurement error to a third of its mixed-frequency value, forces the AR(2) cycle to a near-unit root, and through that opens a near-exact ridge between the trend and cycle states (posterior correlation -0.9996 against $+0.13$). We report it in Section 7 as a comparison rather than as the main specification.
- **Identification caveat.** The variation is concentrated in the 1980s and $\bar{N}_t$ is nearly collinear with a linear trend, so δ is an association, not a causal effect.

11 Status and open items

Estimation status (auto-generated). The main results use the 77 runs of the mixed-frequency design, of which 60 fail the scalar acceptance rule (maximum $\hat{R} \leq 1.01$ and minimum bulk ESS ≥ 400 over every stored scalar). The interpolated comparison of Section 7 uses 77 runs, of which 38 fail the same rule. CES carries no latent firm-count state, so its 16 cells are shared across both designs. All headline estimates, tables, and figures are generated automatically from the saved posteriors; no numbers from earlier revisions are read in. Section 8 states which failures bear on the conclusions and which do not.

Settled in the current revision. The positive δ and flattening κ_t for the CPI negative-unemployment-gap specification of HSA steady, and their insensitivity to the firm-count observation design; the positive sign of the core-CPI HP specification and the non-result of BN; the non-results of the three PPI activity specifications; the prior sensitivity of δ and its decomposition into δ 's own prior versus the AR(2) prior; the economic magnitude; the CES–HSA slope-bias diagnostic via the inverse markup; and the diagnosis that the latent-level ridge is a property of the interpolated design rather than of the data.

Open (empirical / econometric).

- **Causal identification.** The horse race against a linear time trend cannot distinguish the two. Before any claim beyond association, add quadratic trends, structural-break dummies, and placebo interactions.
- **Expectation mismatch.** Beyond core-matched expected inflation, connect a PPI forecast series (e.g. the Livingston Survey) so the expectation term matches the price index.
- **Genuinely out-of-sample evaluation.** Table 6 now reports prequential LPD_1 , WAIC and PSIS-LOO next to the plug-in score, but all four are computed on the estimation sample. A rolling out-of-sample exercise—re-estimating on an expanding window and scoring only genuinely future quarters—remains open, and would be the only way to settle whether the time-varying slope forecasts better or merely fits better. The PPI and HSA-full cells additionally have maximum Pareto $\hat{k} > 0.7$, so their LOO values should be replaced by exact or K -fold cross-validation.
- **Period robustness.** Re-estimate the current revision using only contiguous quarters. The pre-2008/post-1988 numbers from earlier revisions are not used for the conclusions of this paper.
- **Bayes-factor robustness.** In addition to the two-sided SDDR, report the one-sided ($\delta > 0$) evaluation and the KDE bandwidth sensitivity.
- **Annual-Q4 state block.** The marginal chains of δ, κ_0 are good, but ρ_1, ρ_2 and some time points of $\bar{N}_t, \hat{N}_t$ have low ESS / high Rhats. Consider reparameterization, longer independent chains, and priors that relax the weak identification of the state equation.

Open (computational).

- **Conditional marginal likelihood.** A corrected implementation of $p(\pi \mid x, N^{\text{obs}}, M)$ now exists: the latent states are integrated out exactly by the Kalman filter through $\log p(\pi \mid N^{\text{obs}}, \theta) = \log p(\pi, N^{\text{obs}} \mid \theta) - \log p(N^{\text{obs}} \mid \theta)$, the filter uses the estimating model’s own initial state (earlier versions started the trend at N_0^{obs} , evaluating a different model and using the first observation twice), the AR(2) ordinate is the sampler’s own initial-lag conditional, the truncated (ρ_1, ρ_2) prior and conditional are normalized over the stationary triangle, and the ordinate uses Chib’s sequential factorization with explicit reduced Gibbs runs. HSA full is excluded by construction: its bilinear latent term is not linear-Gaussian, and the previous “conditional” figure for it conditioned on posterior-*mean* states, which is a plug-in and not a marginal likelihood. These numbers are computed but not yet promoted into any table.
- **Weak convergence of full.** HSA full remains outside the criteria in every cell under Particle Gibbs. HSA const-theta’s failure, by contrast, was a blocking problem and is resolved by the exact joint FFBS; that is a computational fix and is not evidence about identification.
- **Weak identification of γ .** An identification strategy is needed to relax the collinearity between $\hat{N}\bar{N}$ and $\hat{N}$.
- **Latent-level ridge.** $\bar{N}_t$ and $\hat{N}_t$ have posterior correlation near -0.999 ; only their sum is pinned by the firm-count equation. κ_0 is therefore identified partly through the initial-state prior, which is held fixed across the prior sweep. δ , which loads on the interaction $x_t\bar{N}_t$, is not affected by this ridge.

A Appendix: results by output gap (mixed-frequency, main design)

The body takes the negative unemployment gap as the main activity measure and separates the output-gap results here. Because HP and BN estimate potential output differently, we do not combine them into a single “output gap” coefficient but report the full baseline results by price index (3) and model (5) for each definition. The slope is the constant κ for CES/HSA dynamic and the slope at average competition κ_0 for the other HSA models. δ is the sensitivity of the slope to the firm-count trend, and entry denotes θ or θ_0 .

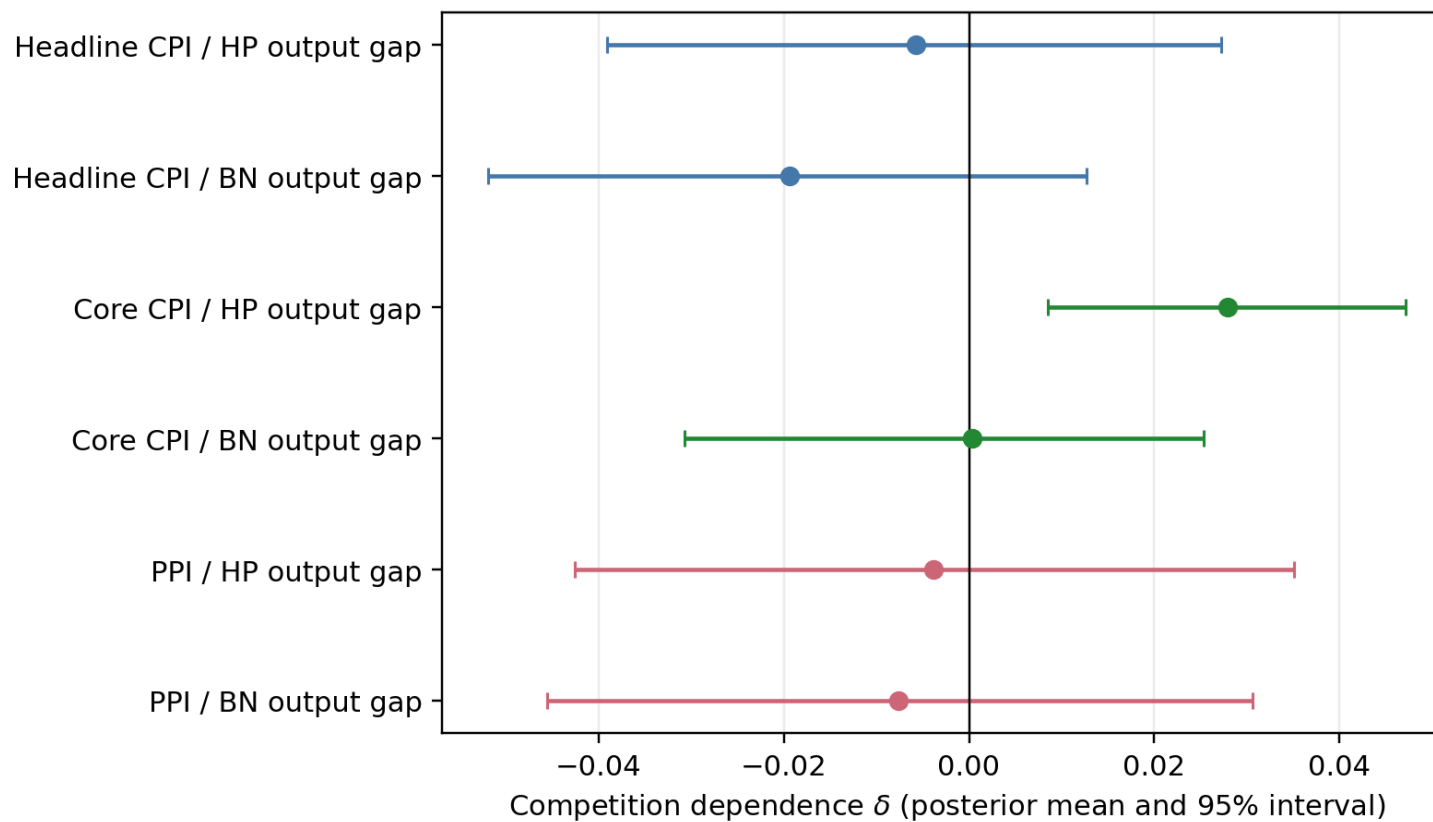


Figure 7: HSA-steady δ by output gap (posterior mean and 95% CI). HP and BN are separated and headline CPI, core CPI, and PPI are shown on the same scale.

A.1 HP output gap

For HSA steady the headline-CPI δ is near zero, while core CPI is positive:

$$\delta_{\text{Core,HP}} = +0.028[+0.009, +0.047], \quad \text{BF}_{10} = 19.6.$$

The PPI $\delta = -0.004[-0.043, +0.035]$ includes zero. The positive competition dependence of the HP specification is therefore limited to core CPI and is not a result for price indices in general.

Table 15: HP output gap: baseline estimates by price index and model. † marks failure of the convergence criteria.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.140 [+0.050, +0.232]	–	–	–	–	–	OK
CES	Core CPI	+0.091 [+0.042, +0.140]	–	–	–	–	–	OK
CES	PPI	+0.210 [-0.062, +0.470]	–	–	–	–	–	OK
HSA steady	Headline CPI	+0.140 [+0.048, +0.231]†	-0.006 [-0.039, +0.027]†	0.9	–	–	+0.135 → +0.155	watch
HSA steady	Core CPI	+0.073 [+0.022, +0.124]†	+0.028 [+0.009, +0.047]†	19.6	–	–	+0.152 → +0.001	watch
HSA steady	PPI	+0.212 [-0.040, +0.469]†	-0.004 [-0.043, +0.035]†	1.0	–	–	+0.207 → +0.221	watch
HSA dynamic	Headline CPI	+0.139 [+0.052, +0.226]†	–	–	+0.054 [-0.169, +0.346]†	–	–	watch
HSA dynamic	Core CPI	+0.090 [+0.046, +0.135]†	–	–	+0.014 [-0.183, +0.320]†	–	–	watch
HSA dynamic	PPI	+0.162 [-0.109, +0.445]†	–	–	+0.112 [-0.215, +0.454]†	–	–	watch
HSA const-theta	Headline CPI	+0.140 [+0.050, +0.233]†	-0.006 [-0.040, +0.028]†	0.9	+0.052 [-0.187, +0.350]†	–	+0.136 → +0.154	watch
HSA const-theta	Core CPI	+0.075 [+0.022, +0.125]†	+0.027 [+0.004, +0.048]†	12.7	-0.027 [-0.106, +0.046]†	–	+0.146 → +0.005	watch
HSA const-theta	PPI	+0.212 [-0.053, +0.492]†	-0.003 [-0.042, +0.036]†	1.0	+0.109 [-0.245, +0.457]†	–	+0.207 → +0.221	watch
HSA full	Headline CPI	+0.138 [+0.047, +0.232]†	-0.007 [-0.041, +0.026]†	0.9	+0.065 [-0.205, +0.376]†	-0.001 [-0.039, +0.039]†	+0.136 → +0.156	watch
HSA full	Core CPI	+0.072 [+0.019, +0.126]†	+0.026 [+0.003, +0.047]†	8.3	-0.015 [-0.143, +0.099]†	-0.006 [-0.040, +0.029]†	+0.144 → +0.005	watch
HSA full	PPI	+0.209 [-0.061, +0.471]†	-0.004 [-0.043, +0.035]†	1.0	+0.112 [-0.229, +0.465]†	+0.000 [-0.038, +0.040]†	+0.203 → +0.220	watch

A.2 BN output gap

For HSA steady the δ point estimates are negative for headline CPI, positive for core CPI, and negative for PPI, but all three 95% credible intervals include zero (headline $-0.019[-0.052, +0.013]$, core $+0.000[-0.031, +0.025]$, PPI $-0.008[-0.046, +0.031]$). Each BF_{10} is also about one, so the BN specification provides no evidence for a competition dependence of the slope.

Table 16: BN output gap: baseline estimates by price index and model. † marks failure of the convergence criteria.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.059 [-0.025, +0.144]	—	—	—	—	—	OK
CES	Core CPI	+0.032 [-0.018, +0.082]	—	—	—	—	—	OK
CES	PPI	+0.189 [-0.049, +0.428]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.053 [-0.033, +0.139] [†]	-0.019 [-0.052, +0.013] [†]	1.6	—	—	+0.037 → +0.103	watch
HSA steady	Core CPI	+0.030 [-0.021, +0.081] [†]	+0.000 [-0.031, +0.025] [†]	0.7	—	—	+0.041 → +0.030	watch
HSA steady	PPI	+0.187 [-0.051, +0.437] [†]	-0.008 [-0.046, +0.031] [†]	1.0	—	—	+0.177 → +0.207	watch
HSA dynamic	Headline CPI	+0.059 [-0.025, +0.142] [†]	—	—	+0.055 [-0.158, +0.323] [†]	—	—	watch
HSA dynamic	Core CPI	+0.030 [-0.015, +0.077] [†]	—	—	-0.019 [-0.163, +0.262] [†]	—	—	watch
HSA dynamic	PPI	+0.156 [-0.091, +0.400] [†]	—	—	+0.117 [-0.227, +0.466] [†]	—	—	watch
HSA const-theta	Headline CPI	+0.051 [-0.037, +0.138] [†]	-0.019 [-0.053, +0.013] [†]	1.6	+0.058 [-0.185, +0.355] [†]	—	+0.036 → +0.102	watch
HSA const-theta	Core CPI	+0.024 [-0.030, +0.075] [†]	-0.007 [-0.037, +0.021] [†]	0.8	+0.021 [-0.191, +0.349] [†]	—	+0.033 → +0.042	watch
HSA const-theta	PPI	+0.190 [-0.051, +0.418] [†]	-0.007 [-0.045, +0.032] [†]	1.1	+0.116 [-0.241, +0.472] [†]	—	+0.179 → +0.209	watch
HSA full	Headline CPI	+0.063 [-0.029, +0.161] [†]	-0.019 [-0.048, +0.011] [†]	1.7	-0.022 [-0.350, +0.348] [†]	-0.008 [-0.054, +0.035] [†]	+0.027 → +0.112	watch
HSA full	Core CPI	+0.029 [-0.025, +0.085] [†]	-0.002 [-0.037, +0.026] [†]	0.9	+0.027 [-0.167, +0.375] [†]	-0.006 [-0.047, +0.034] [†]	+0.046 → +0.034	watch
HSA full	PPI	+0.186 [-0.054, +0.433] [†]	-0.007 [-0.045, +0.031] [†]	1.0	+0.110 [-0.257, +0.485] [†]	+0.000 [-0.038, +0.040] [†]	+0.180 → +0.203	watch

How to read the whole. HP core CPI agrees in direction with the CPI result for the negative unemployment gap, but HP headline, HP PPI, and all price indices of BN do not support it. And because HSA full has every cell, and const-theta most cells, outside the convergence criteria, we do not count sign agreement in the complex models as robustness evidence. The output-gap results are not a claim that “the result always holds across activity measures” but a sensitivity analysis showing which combinations of measurement and price index the result depends on.

B Appendix: results by output gap (interpolated design, for comparison)

We apply the same mixed-frequency estimation as Section 4 to the HP and BN output gaps. Each table gives the full baseline results by price index (3) and model (5); only the CES row is a common reference that does not depend on the firm-count observation scheme. We first compare the HSA-steady δ on the same scale, then show the full coefficients and diagnostics by model.

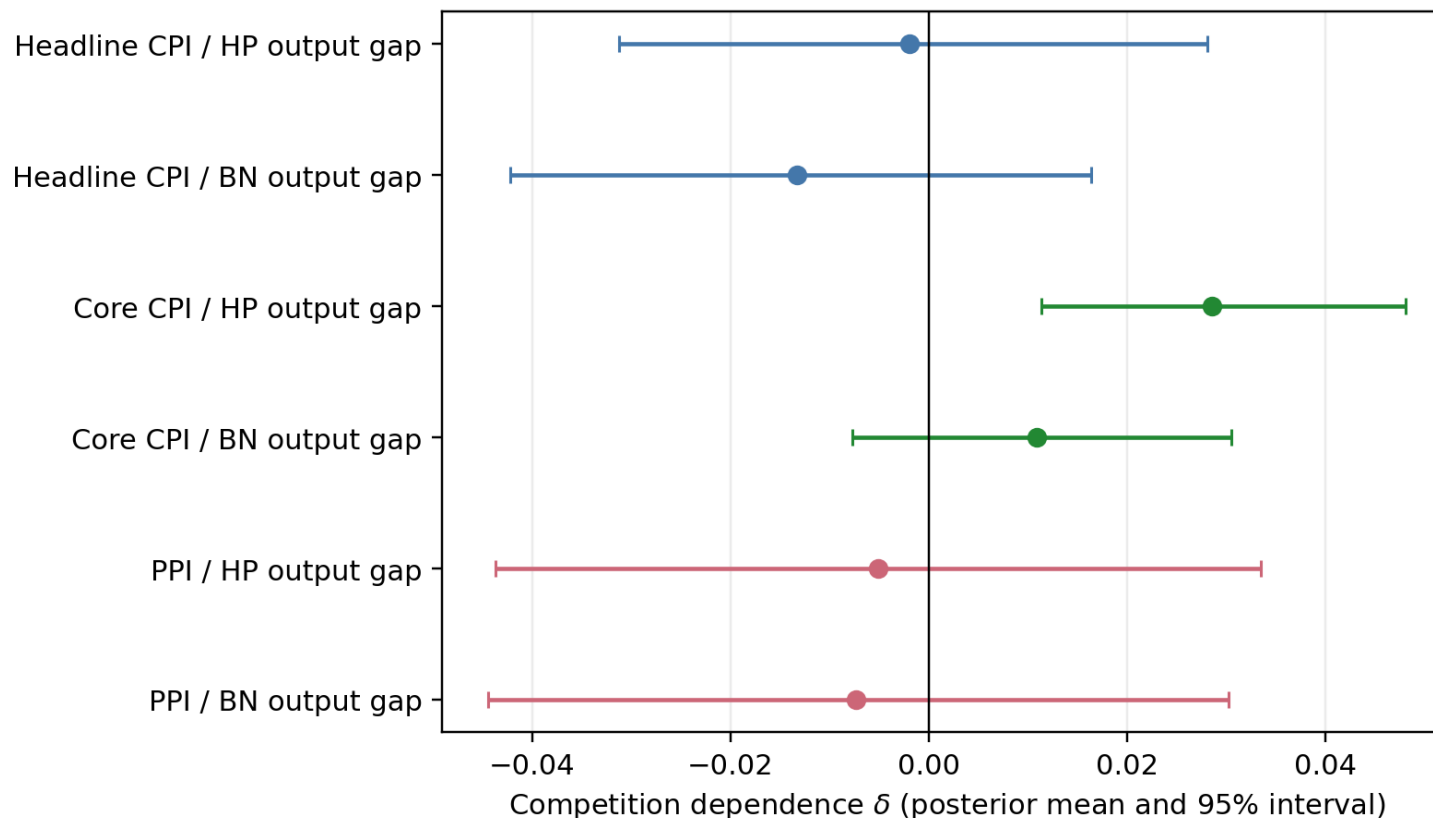


Figure 8: Interpolated design: HSA-steady δ by HP and BN output gap (posterior mean and 95% CI).

B.1 HP output gap

For the annual-Q4 HSA steady, headline CPI gives $\delta = -0.006[-0.039, +0.027]$ ($\text{BF}_{10} = 0.9$), core CPI gives $\delta = +0.028[+0.009, +0.047]$ ($\text{BF}_{10} = 19.6$), and PPI gives $\delta = -0.004[-0.043, +0.035]$ ($\text{BF}_{10} = 1.0$).

Table 17: Interpolated design: HP output gap, baseline estimates by price index and model.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.140 [+0.050, +0.232]	—	—	—	—	—	OK
CES	Core CPI	+0.091 [+0.042, +0.140]	—	—	—	—	—	OK
CES	PPI	+0.210 [-0.062, +0.470]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.141 [+0.048, +0.237]	-0.002 [-0.031, +0.028]	0.7	—	—	+0.135 → +0.146	OK
HSA steady	Core CPI	+0.069 [+0.018, +0.123]	+0.029 [+0.011, +0.048]	67.7	—	—	+0.161 → -0.005	OK
HSA steady	PPI	+0.210 [-0.067, +0.473]	-0.005 [-0.044, +0.034]	1.0	—	—	+0.193 → +0.224	OK
HSA dynamic	Headline CPI	+0.130 [+0.038, +0.222] [†]	—	—	-0.026 [-0.123, +0.106] [†]	—	—	watch
HSA dynamic	Core CPI	+0.085 [+0.037, +0.132]	—	—	-0.025 [-0.072, +0.029]	—	—	OK
HSA dynamic	PPI	+0.163 [-0.114, +0.437] [†]	—	—	+0.019 [-0.268, +0.328] [†]	—	—	watch
HSA const-theta	Headline CPI	+0.136 [+0.034, +0.237] [†]	-0.004 [-0.035, +0.029] [†]	0.8	-0.015 [-0.134, +0.126] [†]	—	+0.121 → +0.149	watch
HSA const-theta	Core CPI	+0.067 [+0.013, +0.121]	+0.028 [+0.009, +0.049]	39.6	-0.006 [-0.053, +0.047]	—	+0.157 → -0.004	OK
HSA const-theta	PPI	+0.217 [-0.049, +0.485] [†]	-0.005 [-0.043, +0.034] [†]	1.0	+0.012 [-0.273, +0.343] [†]	—	+0.202 → +0.230	watch
HSA full	Headline CPI	+0.136 [+0.040, +0.233] [†]	-0.006 [-0.037, +0.028] [†]	0.9	+0.029 [-0.100, +0.147] [†]	-0.019 [-0.057, +0.022] [†]	+0.115 → +0.151	watch
HSA full	Core CPI	+0.083 [+0.022, +0.192] [†]	+0.022 [+0.006, +0.052] [†]	15.6	+0.022 [-0.046, +0.092] [†]	-0.014 [-0.045, +0.017] [†]	+0.145 → +0.006	watch
HSA full	PPI	+0.213 [-0.059, +0.488] [†]	-0.004 [-0.044, +0.033] [†]	1.0	+0.018 [-0.271, +0.318] [†]	-0.005 [-0.044, +0.034] [†]	+0.199 → +0.224	watch

B.2 BN output gap

For the annual-Q4 HSA steady, headline CPI gives $\delta = -0.019[-0.052, +0.013]$ ($\text{BF}_{10} = 1.6$), core CPI gives $\delta = +0.000[-0.031, +0.025]$ ($\text{BF}_{10} = 0.7$), and PPI gives $\delta = -0.008[-0.046, +0.031]$ ($\text{BF}_{10} = 1.0$).

Table 18: Interpolated design: BN output gap, baseline estimates by price index and model.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.059 [-0.025, +0.144]	—	—	—	—	—	OK
CES	Core CPI	+0.032 [-0.018, +0.082]	—	—	—	—	—	OK
CES	PPI	+0.189 [-0.049, +0.428]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.064 [-0.024, +0.153]	-0.013 [-0.042, +0.016]	1.1	—	—	+0.022 → +0.097	OK
HSA steady	Core CPI	+0.031 [-0.020, +0.083]	+0.011 [-0.008, +0.031]	0.9	—	—	+0.063 → +0.005	OK
HSA steady	PPI	+0.190 [-0.054, +0.427] [†]	-0.007 [-0.044, +0.030] [†]	1.1	—	—	+0.167 → +0.210	watch
HSA dynamic	Headline CPI	+0.057 [-0.026, +0.137] [†]	—	—	-0.076 [-0.170, +0.083] [†]	—	—	watch
HSA dynamic	Core CPI	+0.034 [-0.012, +0.081]	—	—	-0.040 [-0.087, +0.013]	—	—	OK
HSA dynamic	PPI	+0.159 [-0.098, +0.404] [†]	—	—	-0.040 [-0.325, +0.300] [†]	—	—	watch
HSA const-theta	Headline CPI	+0.063 [-0.025, +0.151] [†]	-0.011 [-0.038, +0.022] [†]	1.0	-0.071 [-0.169, +0.081] [†]	—	+0.019 → +0.097	watch
HSA const-theta	Core CPI	+0.029 [-0.024, +0.079]	+0.014 [-0.004, +0.032]	1.8	-0.049 [-0.098, +0.005]	—	+0.081 → -0.006	OK
HSA const-theta	PPI	+0.191 [-0.053, +0.437] [†]	-0.008 [-0.045, +0.030] [†]	1.1	-0.039 [-0.317, +0.295] [†]	—	+0.166 → +0.211	watch
HSA full	Headline CPI	+0.071 [-0.041, +0.168] [†]	-0.008 [-0.042, +0.036] [†]	1.2	+0.051 [-0.092, +0.170] [†]	-0.016 [-0.062, +0.034] [†]	+0.044 → +0.103	watch
HSA full	Core CPI	+0.046 [-0.004, +0.096] [†]	+0.012 [-0.005, +0.030] [†]	1.2	+0.024 [-0.061, +0.110] [†]	-0.033 [-0.064, -0.003] [†]	+0.086 → +0.014	watch
HSA full	PPI	+0.198 [-0.039, +0.444] [†]	-0.006 [-0.045, +0.033] [†]	1.0	+0.052 [-0.261, +0.378] [†]	-0.005 [-0.043, +0.034] [†]	+0.184 → +0.212	watch

These correspond one-to-one with Appendix A of the PCHIP version. In the comparison we read not only the point estimate but the 95% credible interval, the Bayes factor, and the convergence flag together. Where an interval widens under annual-Q4, that is the result of not treating the quarterly firm count as observed and propagating the interpolation uncertainty.

C Appendix: full model-by-model tables (negative unemployment gap)

This appendix collects the dense, auto-generated tables that support Sections 5 and 9: the full model-by-model coefficients for the negative unemployment gap and the whole-run convergence tally. (The three-prior sensitivity table now sits with the robustness discussion, Table 14.) The slope is κ for CES/HSA dynamic and κ_0 for the other models; entry is θ or θ_0 ; † marks cells outside the convergence criteria.

Table 19: Model-by-model results for the negative-unemployment-gap specification. Slope is κ for CES/HSA dynamic and κ_0 otherwise; entry is θ or θ_0 . † marks cells that fail the convergence criteria and are not interpreted.

model	inflation	slope	delta	BF10(delta)	entry	gamma	kappa path	diagnostics
CES	Headline CPI	+0.057 [-0.006, +0.120]	—	—	—	—	—	OK
CES	Core CPI	+0.040 [+0.007, +0.071]	—	—	—	—	—	OK
CES	PPI	+0.073 [-0.145, +0.284]	—	—	—	—	—	OK
HSA steady	Headline CPI	+0.075 [+0.009, +0.145]	+0.029 [+0.005, +0.053]	11.2	—	—	+0.143 → -0.001	watch
HSA steady	Core CPI	+0.056 [+0.021, +0.092]	+0.022 [+0.008, +0.038]	36.2	—	—	+0.110 → -0.002	watch
HSA steady	PPI	+0.080 [-0.134, +0.296]	+0.009 [-0.028, +0.047]	1.1	—	—	+0.093 → +0.055	watch
HSA dynamic	Headline CPI	+0.055 [-0.011, +0.117]	—	—	+0.037 [-0.211, +0.319]	—	—	watch
HSA dynamic	Core CPI	+0.040 [+0.009, +0.071]	—	—	-0.012 [-0.173, +0.289]	—	—	watch
HSA dynamic	PPI	+0.051 [-0.155, +0.263]	—	—	+0.101 [-0.233, +0.431]	—	—	watch
HSA const-theta	Headline CPI	+0.079 [+0.011, +0.152]	+0.029 [+0.003, +0.055]	7.6	+0.022 [-0.159, +0.241]	—	+0.131 → +0.002	watch
HSA const-theta	Core CPI	+0.057 [+0.020, +0.095]	+0.022 [+0.004, +0.038]	21.9	-0.039 [-0.136, +0.054]	—	+0.107 → -0.000	watch
HSA const-theta	PPI	+0.079 [-0.129, +0.289]	+0.009 [-0.030, +0.047]	1.1	+0.108 [-0.246, +0.472]	—	+0.092 → +0.056	watch
HSA full	Headline CPI	+0.080 [+0.011, +0.152]	+0.029 [+0.002, +0.054]	6.8	+0.032 [-0.152, +0.287]	-0.003 [-0.040, +0.037]	+0.129 → +0.005	watch
HSA full	Core CPI	+0.064 [+0.022, +0.109]	+0.025 [+0.008, +0.043]	19.6	+0.041 [-0.134, +0.191]	-0.009 [-0.042, +0.027]	+0.140 → -0.003	watch
HSA full	PPI	+0.075 [-0.140, +0.284]	+0.010 [-0.029, +0.047]	1.1	+0.112 [-0.238, +0.446]	+0.000 [-0.038, +0.039]	+0.091 → +0.050	watch

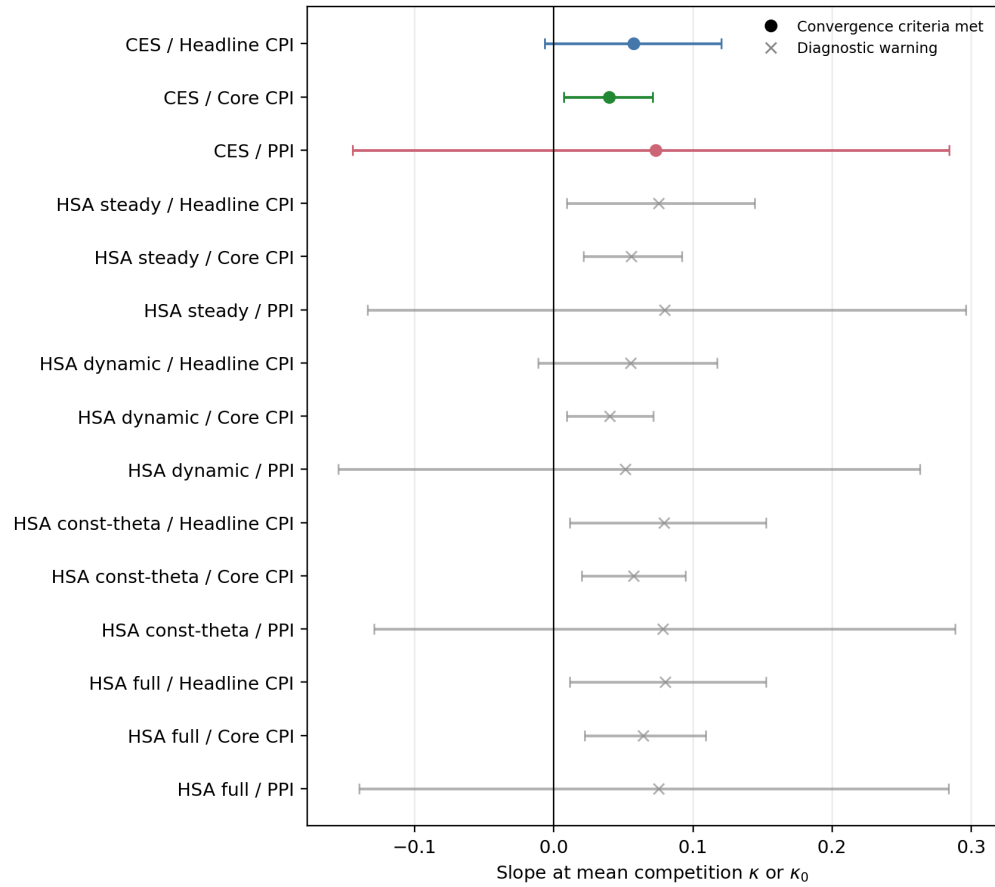


Figure 9: Slope at average competition for the negative-unemployment-gap specification. κ for CES/HSA dynamic and κ_0 otherwise. Circles are runs meeting the convergence criteria; gray crosses are runs to watch.

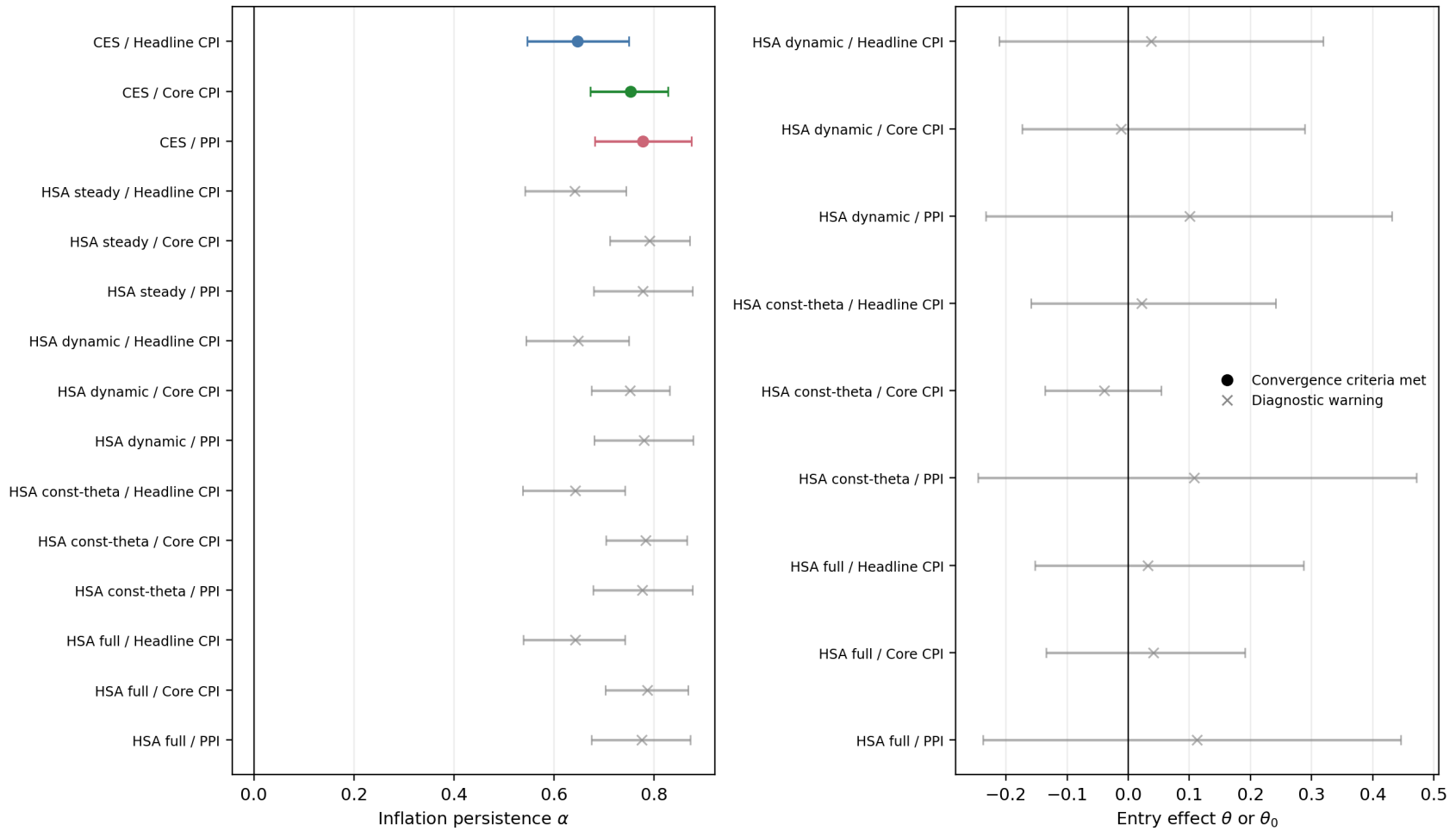


Figure 10: Inflation inertia α (left) and entry coefficient θ or θ_0 (right) for the negative-unemployment-gap specification. Circles and gray crosses are as in Figure 9.

Table 20: Model-by-model convergence tally across all 77 runs. “warnings” counts runs failing the *scalar* rule, “state warnings” the latent-path rule, and “joint warnings” at least one of the three groups (Section 3). The state-block column names the sampler that drew $\bar{N}_t, \hat{N}_t$.

model	sampler	runs	warnings	state warnings	joint warnings	max Rhat	min bulk ESS
CES	conjugate Gibbs	16	0	0	0	1.002	2622
HSA const-theta	joint FFBS	15	15	15	15	1.514	4
HSA dynamic	joint FFBS	16	15	15	15	1.213	7
HSA full	Particle Gibbs	15	15	15	15	1.837	3
HSA steady	joint FFBS	15	15	15	15	1.749	3

Table 21: Group-by-group convergence diagnostics for the negative-unemployment-gap baseline cells: scalar parameters, the trend drift n on its own, the latent-state paths $\bar{N}_t, \hat{N}_t$, and the derived paths κ_t, θ_t . “worst scalar” names the scalar with the lowest bulk ESS. Thresholds are $\hat{R} \leq 1.01$ and bulk ESS ≥ 400 . Mixed-frequency (main) design. This table is the evidence behind the † and “OK (coef)” marks elsewhere: it shows whether a failure is driven by a coefficient of interest or by a nuisance term. Here the worst-mixing scalar is the AR(2) block in almost every cell, which for HSA steady is a nuisance object.

model	inflation	sampler	scalar Rhat	scalar ESS	worst scalar	n Rhat	n ESS	state Rhat	state ESS	derived Rhat	derived ESS	joint
CES	Headline CPI	conjugate Gibbs	1.001	3049	phi_1	–	–	–	–	–	–	OK
CES	Core CPI	conjugate Gibbs	1.001	2905	phi_1	–	–	–	–	–	–	OK
CES	PPI	conjugate Gibbs	1.002	3023	sigma_e	–	–	–	–	–	–	OK
HSA steady	Headline CPI	joint FFBS	1.053	28	rho_1	1.003	290	1.065	39	1.003	269	watch
HSA steady	Core CPI	joint FFBS	1.024	62	rho_1	1.002	485	1.036	67	1.006	153	watch
HSA steady	PPI	joint FFBS	1.016	71	rho_2	1.006	298	1.016	80	1.003	2892	watch
HSA dynamic	Headline CPI	joint FFBS	1.015	68	rho_2	1.004	521	1.025	86	–	–	watch
HSA dynamic	Core CPI	joint FFBS	1.034	57	rho_2	1.004	406	1.024	74	–	–	watch
HSA dynamic	PPI	joint FFBS	1.047	76	rho_2	1.001	312	1.035	61	–	–	watch
HSA const-theta	Headline CPI	joint FFBS	1.024	84	rho_2	1.005	384	1.025	87	1.007	450	watch
HSA const-theta	Core CPI	joint FFBS	1.011	97	sigma_u	1.004	445	1.014	97	1.004	233	watch
HSA const-theta	PPI	joint FFBS	1.021	63	rho_2	1.002	458	1.033	85	1.000	3112	watch
HSA full	Headline CPI	Particle Gibbs	1.128	10	rho_1	1.037	53	1.176	8	1.032	76	watch
HSA full	Core CPI	Particle Gibbs	1.783	3	rho_1	1.084	17	1.673	3	1.266	6	watch
HSA full	PPI	Particle Gibbs	1.117	13	rho_1	1.013	219	1.124	11	1.004	2954	watch